Nearshore wave propagation

MEC_5EO04_TA/MEC_53456_EP – Class 3
October 10, 2025

Marissa L. Yates

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Course schedule

I. Wave dynamics	Project	Lecturer	Date
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	26/09
2. Linear wave theory	Characterizing offshore waves	M. Yates	03/10
3. Nearshore wave propagation	**Calculating extreme wave conditions**	**M. Yates**	**10/10**
4. Coastal hydrodynamics	**Student mini-presentations**	M. Yates	17/10
II. Numerical wave modeling			
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	24/10
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	07/11
III. Wave-structure interactions			
7. Dynamics of a body in waves	Wave load estimation	J. Harris	14/11
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	28/11
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	05/12
10. Final project presentations 13h30 - 17h15			***19 /12***

Nearshore wave propagation

Objectives:

- Review the linear dispersion relation
- Apply wave energy (and action) conservation to estimate:
  - Bathymetric refraction
  - Basic wave-current interactions
  - Diffraction and reflection

Linear wave theory

Linear wave theory: what assumptions do we make?

- **Small amplitude waves**

  H/L<<1

  H/h<<1

- **Flat bottom**

Progressive wave:

$$\eta = a\cos(kx - \omega t)$$

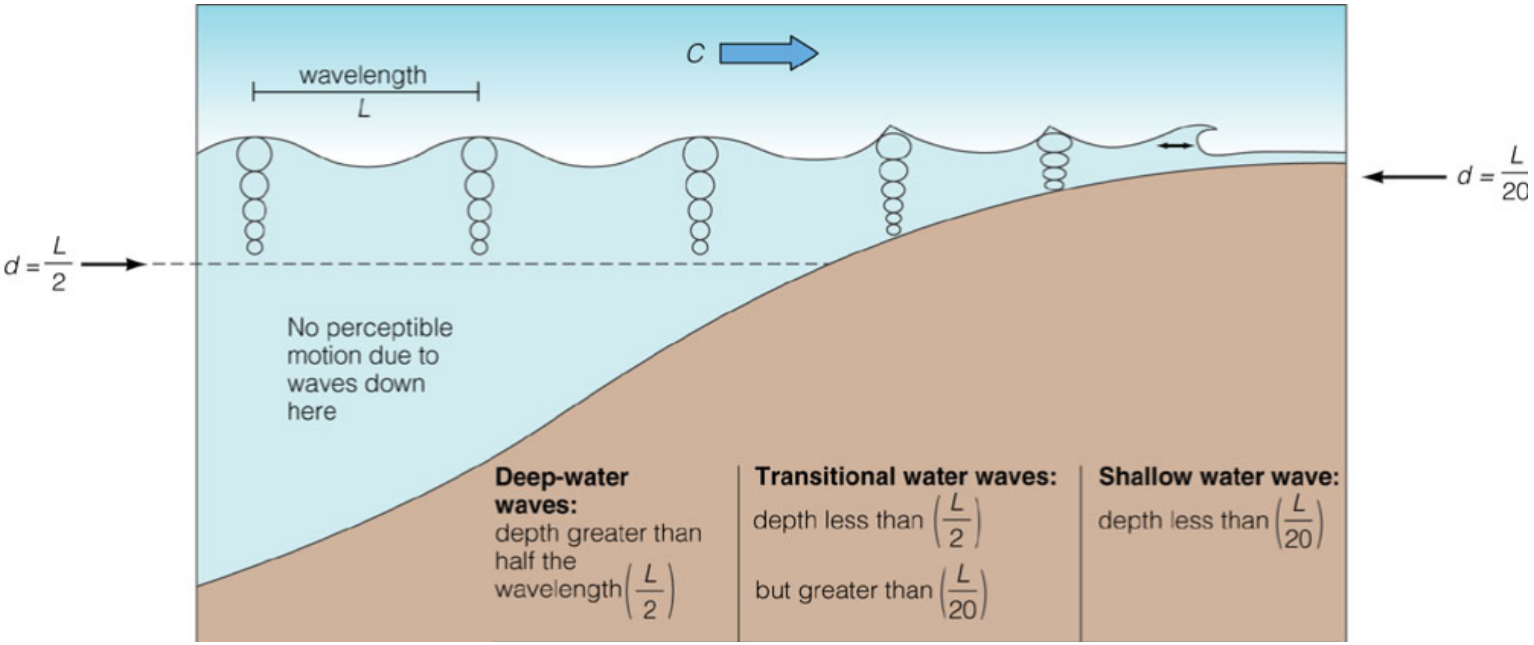
$$\phi = \frac{ag}{\omega}\frac{\cosh(k(z+h))}{\cosh(kh)}\sin(kx - \omega t)$$

Dispersion relation

What is the importance of the dispersion relation?

$$\omega^2 = gk \tanh(kh)$$

Deep and shallow water limits:



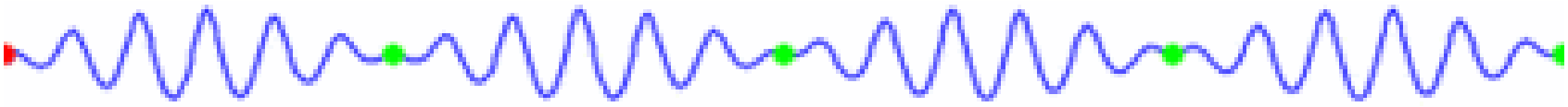
http://science.kennesaw.edu/~jdirnber/oceanography/LecturesOceanogr/LecWaves/

Wave energy and group velocity

- Wave energy: $E = \frac{\rho g H^2}{8}$
- Wave group speed (average speed of wave energy propagation, where $\frac{1}{2} \leq n \leq 1$, such that $Cg \leq C$):

$$C_g = \frac{F_E}{E} = \underbrace{\left[\frac{1}{2}\left(1 + \frac{2kh}{\sinh(2kh)}\right)\right]}_{n} \underbrace{C}_{C} = nC$$

Wave energy propagates slower than individual waves.



https://en.wikipedia.org/wiki/Dispersion_%28water_waves%29#/media/File:Wave_group.gif

Shallow water and deep water approximations for linear wave theory

Variable	Approximations with criteria	
	Shallow-water or long wave approximation (*small kh*)	Deep-water or short wave approximation (*large kh*)
	h/*L* < 1/25 or $T\sqrt{g/h} > 25$	*h*/*L* > 1/2 or $T\sqrt{g/h} < 4$
Dispersion relation	$\omega^2 = gh\,k^2$	$\omega^2 = gk_o$
Wave number *k* (rad/m)	$k = \omega/\sqrt{gh}$	$k_o = \omega^2/g$
Wavelength *L* (m)	$L = T\sqrt{gh}$	$L_o = gT^2/(2\pi)$
Phase speed *c* (m/s)	$c = \sqrt{gh}$	$c_o = gT/(2\pi)$
Group velocity c_g (m/s)	$c_g = c = \sqrt{gh}$	$c_{go} = ½\,c_o = gT/(4\pi)$

Approximations in shallow or deep water

Relative water depth: kh or h/L

Nondimensional wave period: $T\sqrt{gh}$

Variable	Approximations with criteria
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Independent of the water depth

Approximations in shallow or deep water

Relative water depth: kh or h/L

Nondimensional wave period: $T\sqrt{gh}$

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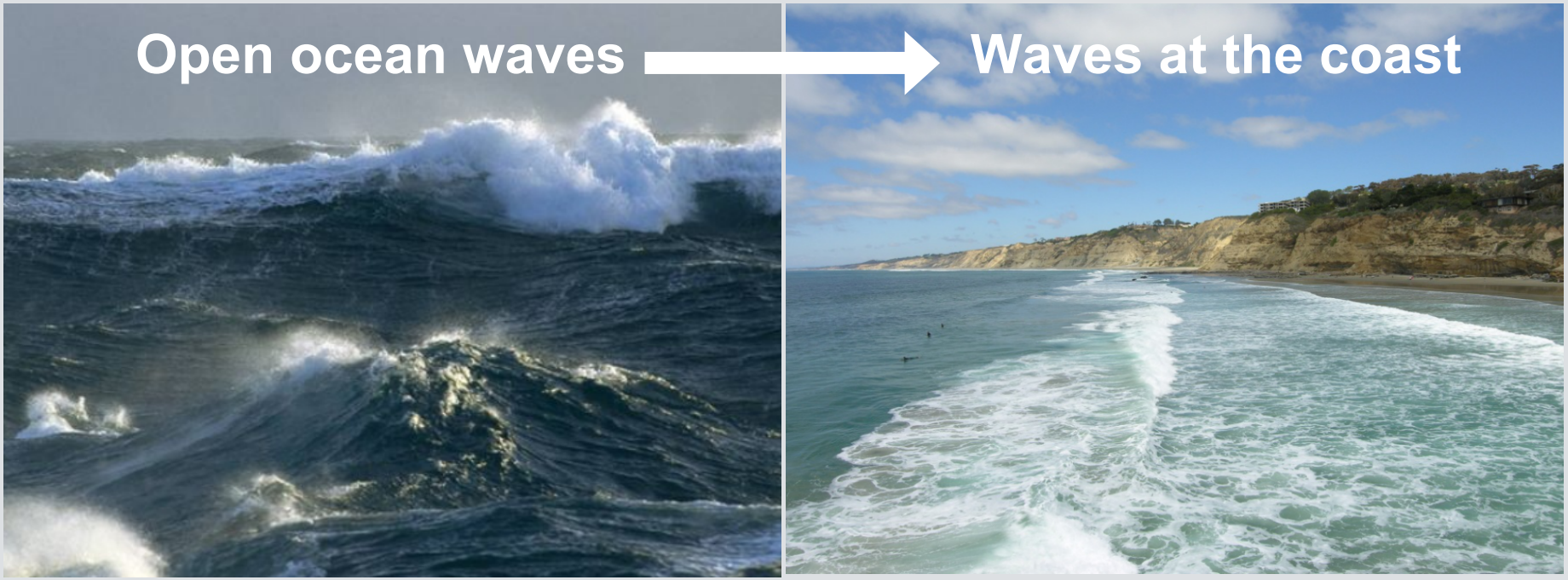
Group velocity c_g (m/s)	$c_g = c = \sqrt{gh}$

Wave phase speed depends only on the water depth

Important physical processes

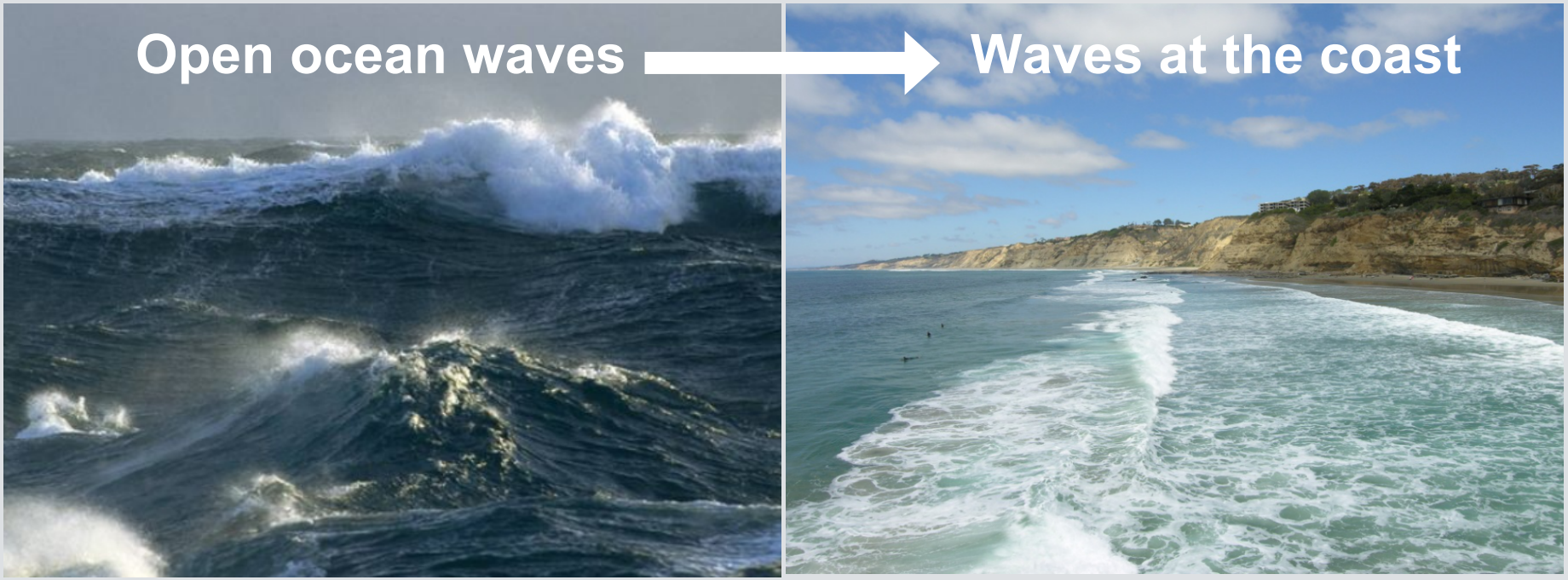


Generation

Propagation and interactions

Dissipation

Important physical processes



Generation
- *by the wind*

Propagation and interactions
- *shoaling*
- *refraction*
- *interactions with currents*
- *reflection*
- *diffraction*
- *interactions between waves*

Dissipation
- *bottom friction*
- *white capping*
- *breaking*

Important physical processes

Physical processes	Ocean	Continental seas	Coastal zone	Ports
Wind generation	+++	+++	+	
Shoaling		++	+++	++
Bathymetric refraction		++	+++	++
Refraction by currents		+	++	
Diffraction			+	+++
Reflection			+	+++
Triplet interactions			++	+
Quadruplet interactions	+++	+++	+	
Bottom friction		+++	++	
White capping	+++	+++	+	
Bathymetric breaking		+	+++	

(adapted from Battjes, 1994)

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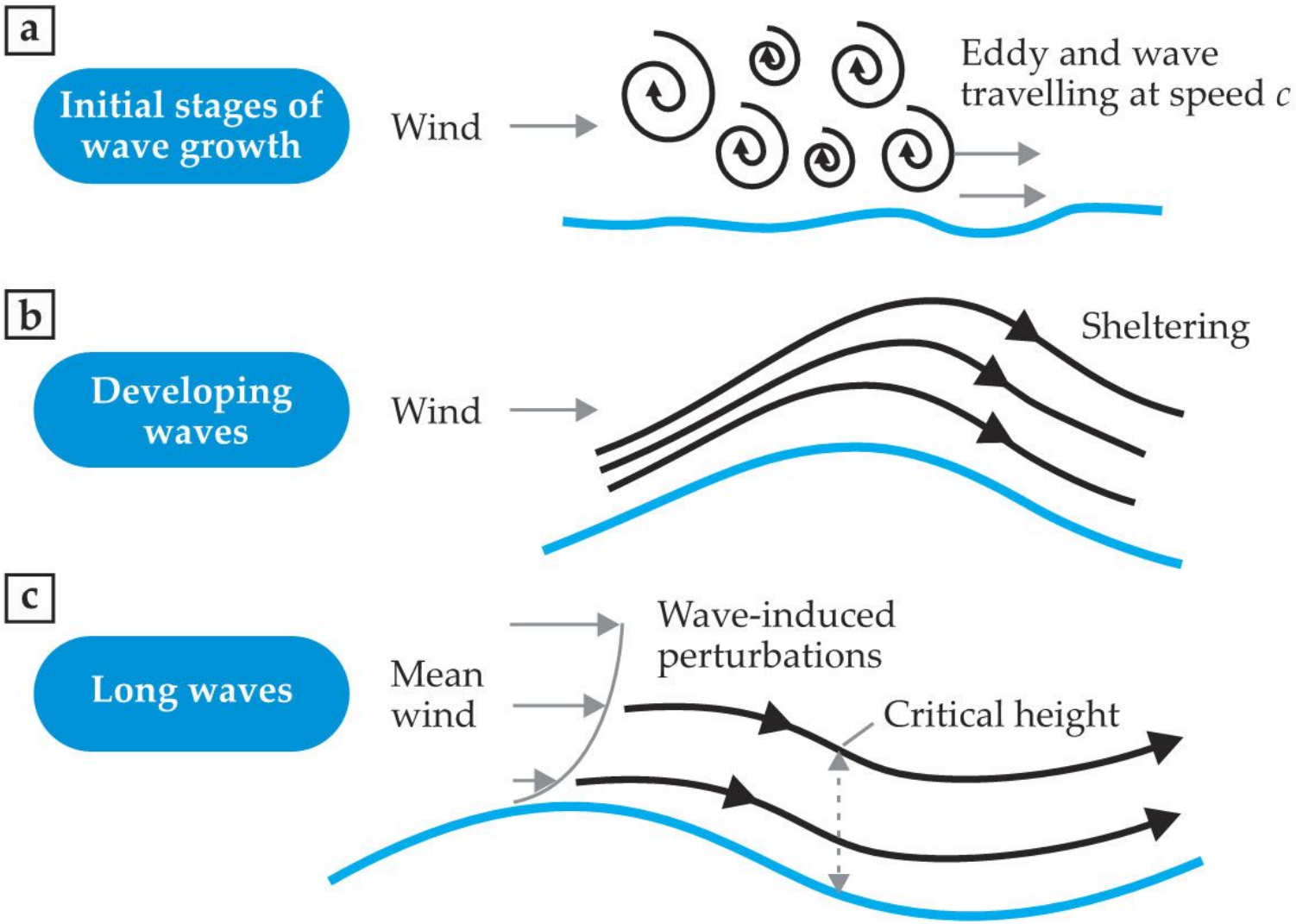
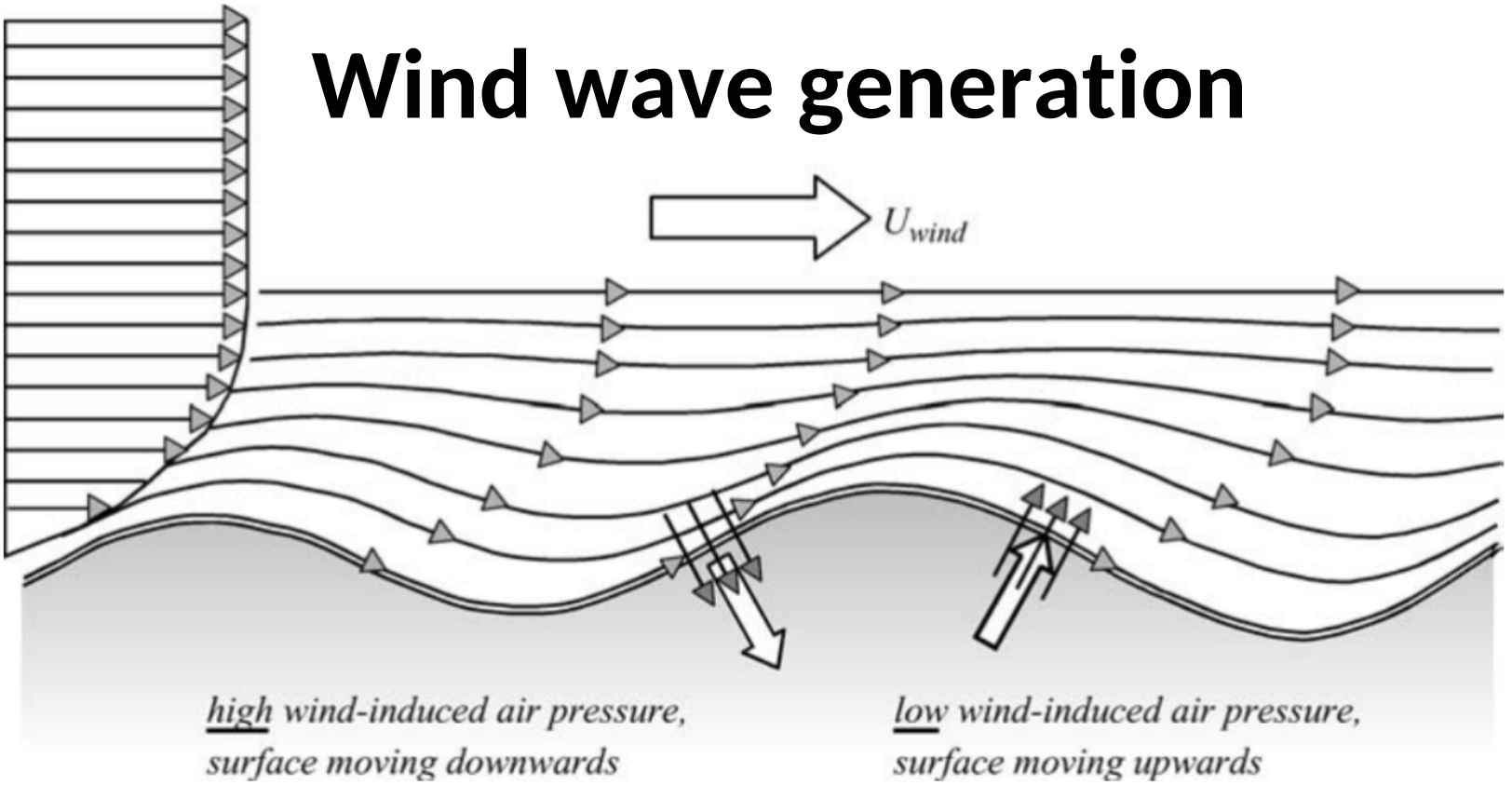
Wind wave generation



https://pubs.aip.org/view-large/figure/47195596/38_1_f2.jpg, accessed 10/2025

Wind wave generation



- Wave-induced wind pressure variation over a propagating wave
- Out-of-phase coupling between pressure and the surface motion => energy transfer to waves(positive feedback)

Holthuijsen, 2007)

Propagation and interactions

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(adapted from Battjes, 1994)

General coastal zone characteristics

- **Range of spatial scales**
  - **Shoaling zone** => O (10L)
  - **Surf zone** => O (L/2)
  - **Swash zone** => O (L/10)

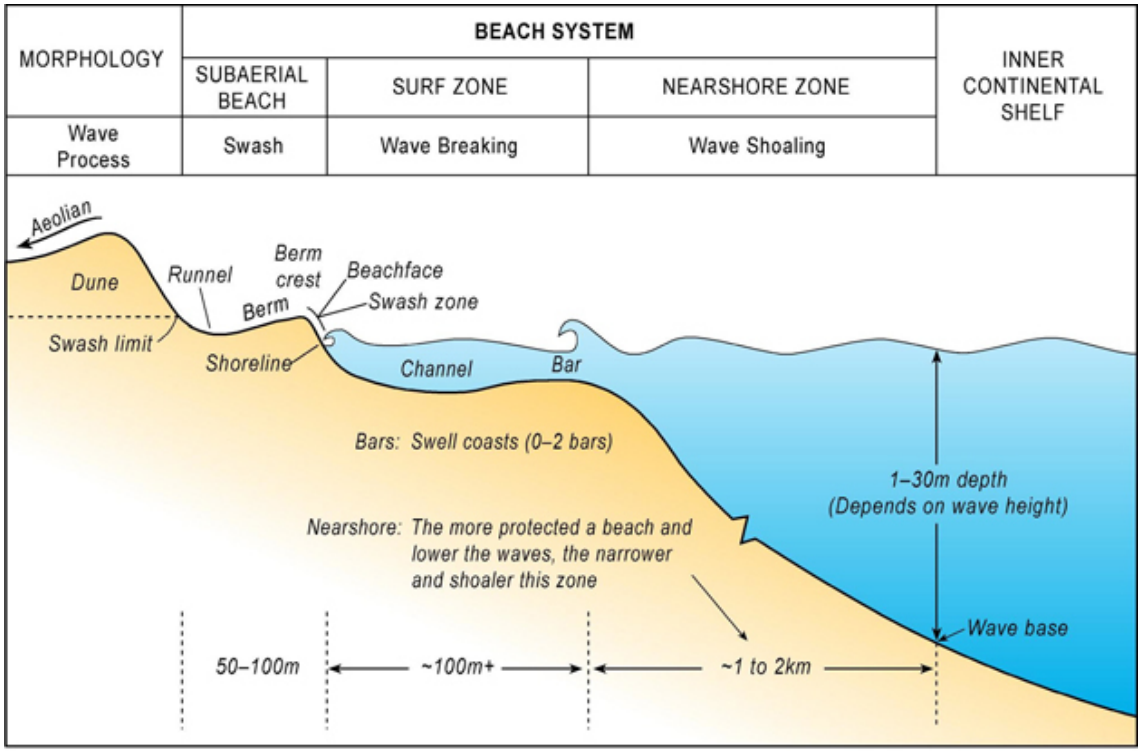


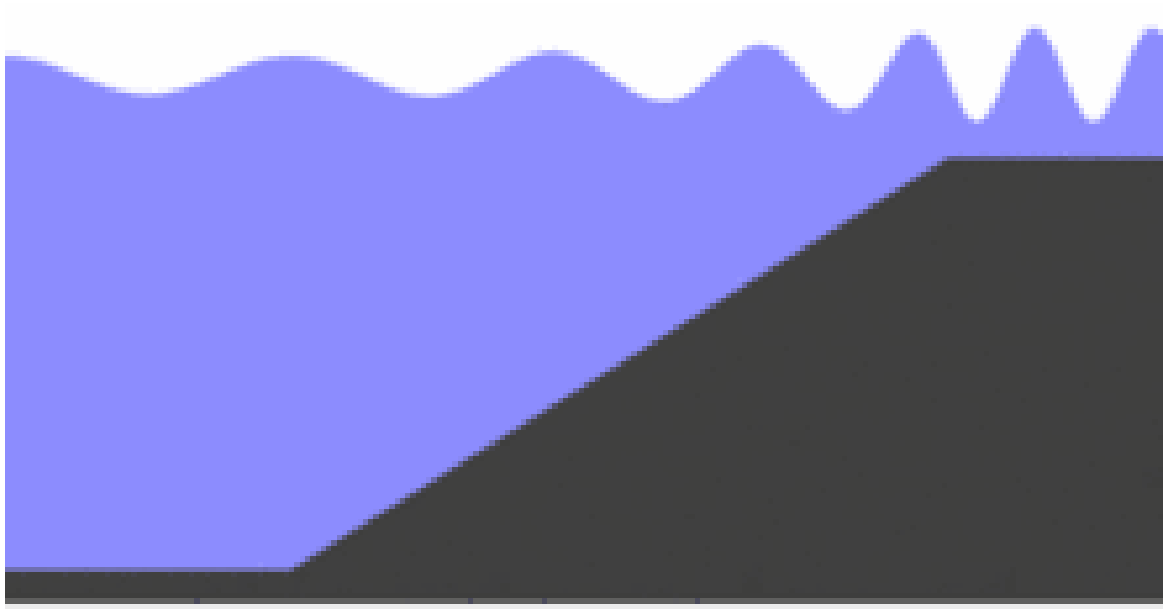
Figure 4.1

Short & Woodroffe 2009

Summary of shoaling

(from the previous lecture…)

Example:



What do you observe as the water depth decreases?

- T, L, C, H?

Summary of shoaling

(from the previous lecture…)

Important results: as waves approach the coastline, if *h* ↘

- **the wavelength *L* ↘**
- **the wavenumber *k* ↗**
- **the wave phase velocity *C* ↘ (=> waves slow down)**

and all of these changes occur monotonically

Important results: as waves approach the coastline, if *h* ↘

- **First, H ↘ until it reaches a minimum of $H=0.913\ H_0$ at $h/L_0=1/(2\pi)$**
- **Then, H ↗ in theory to positive infinity if h->0, but in reality…**

the wave breaks when it reaches the limit of its stability

Propagation and interactions

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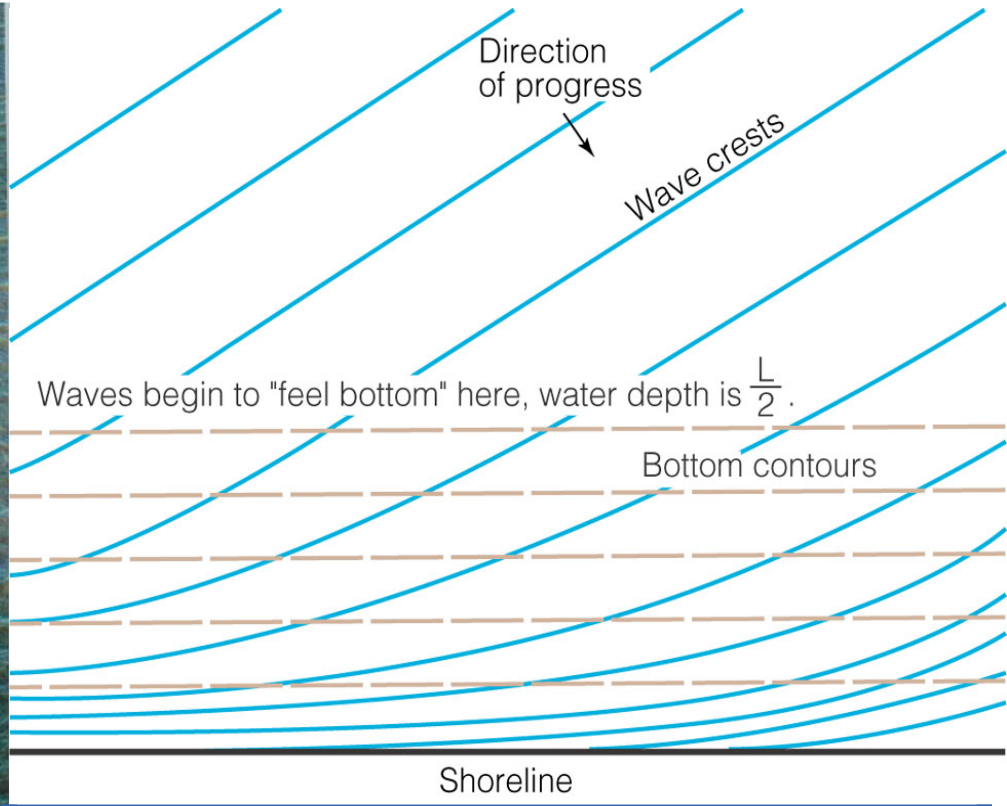
(adapted from Battjes, 1994)

Bathymetric refraction

Whiskey Island, Louisiane, USA
Photo : John Livzey

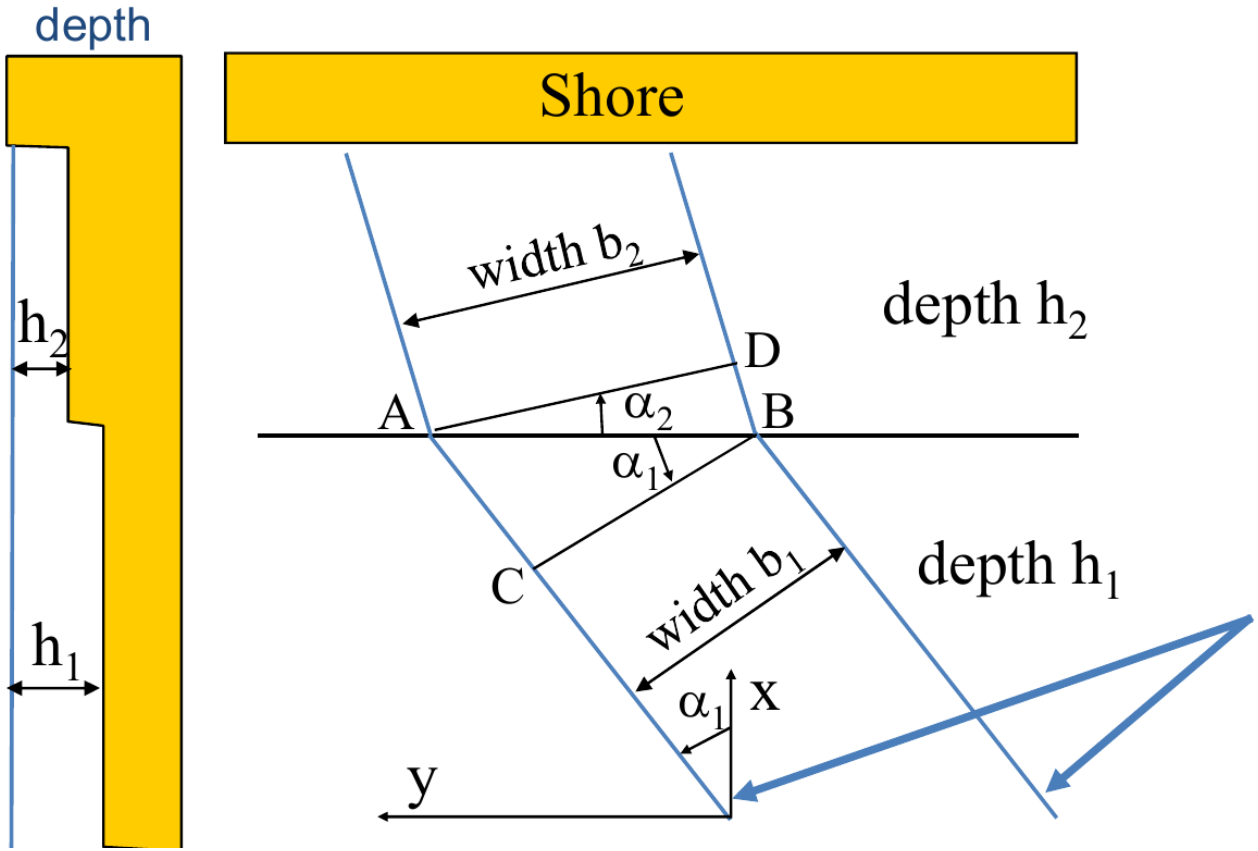


Change in wave propagation direction related to the phase speed

Geometrical refraction

Oblique wave incidence, $\alpha \neq 0$

Start by considering a step change in depth



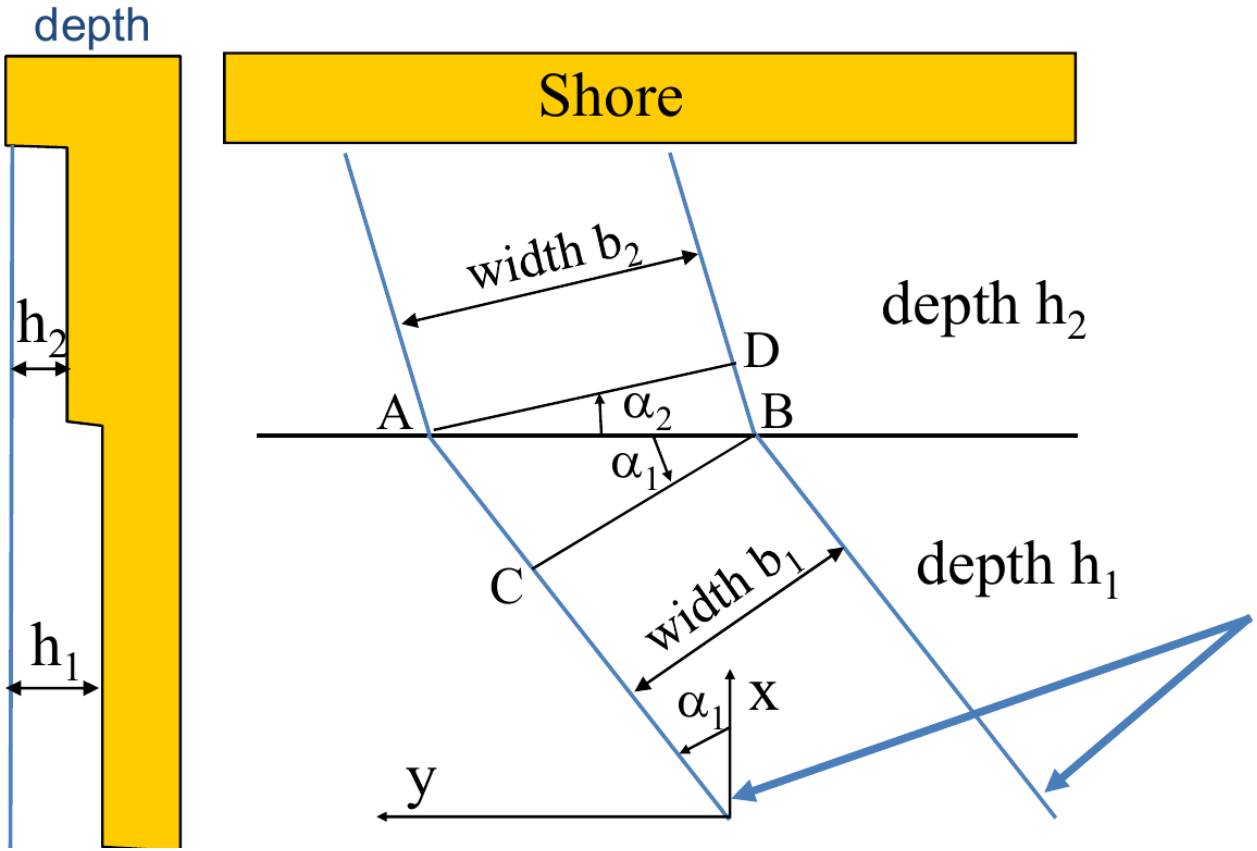
In the shallower region, $h_2 < h_1$, $C_2 < C_1$ since

$$\left.\frac{\partial C}{\partial h}\right|_T > 0$$

Wave orthogonals (lines ⊥ to wave crests)

Geometrical refraction

1. Geometrical relationship: $AB = \frac{b_1}{\cos\alpha_1} = \frac{b_2}{\cos\alpha_2} \rightarrow \frac{b}{\cos\alpha} = c$



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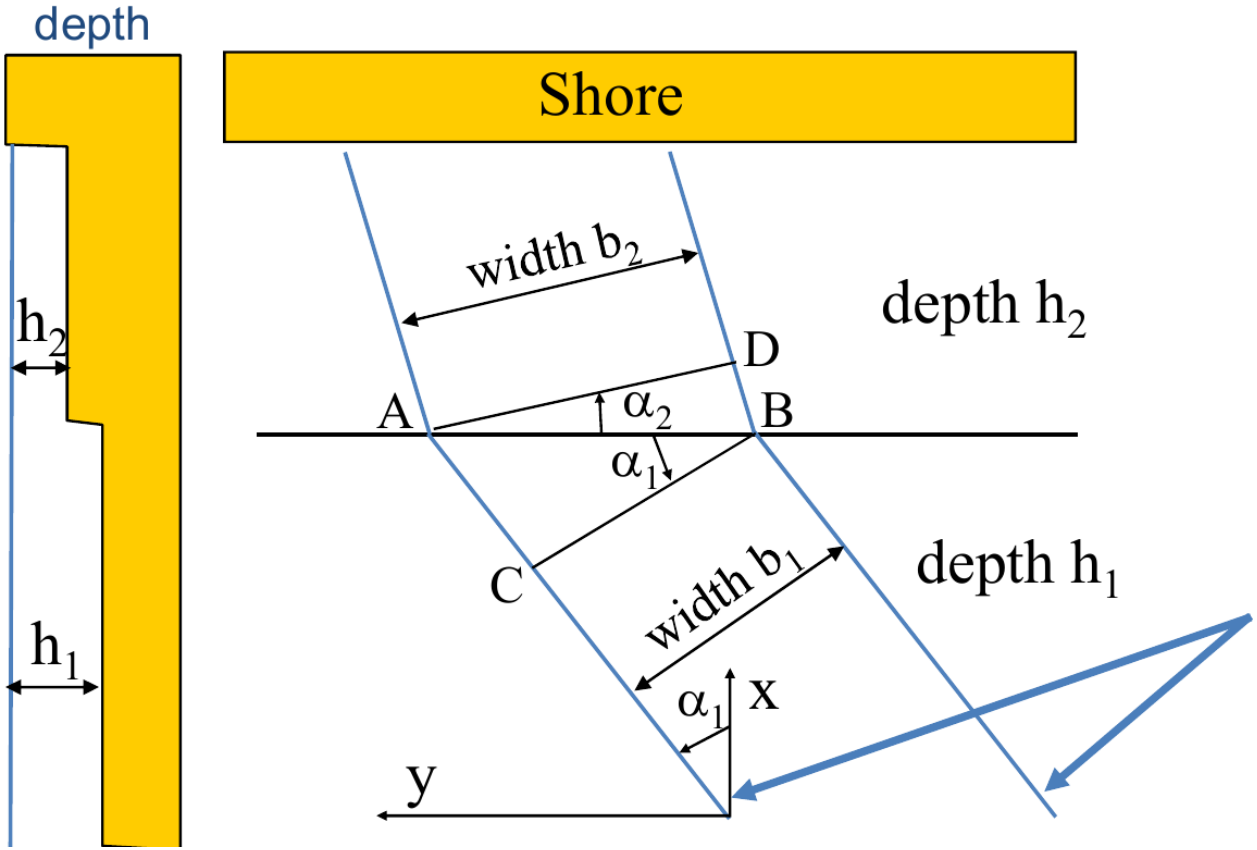
2. Kinematic relationship: same travel time from C->A as B->D



Geometrical refraction

1. Geometrical relationship: $AB = \frac{b_1}{\cos\alpha_1} = \frac{b_2}{\cos\alpha_2} \rightarrow \frac{b}{\cos\alpha} = c$
2. Kinematic relationship: same travel time from C->A as B->D

$$T_{C\to A} = \frac{AC}{C_1} = \frac{AB\sin\alpha_1}{C_1} = T_{B\to D} = \frac{BD}{C_2} = \frac{AB\sin\alpha_2}{C_2}$$

$$\frac{\sin\alpha_1}{C_1} = \frac{\sin\alpha_2}{C_2} \rightarrow \frac{\sin\alpha}{C} = cte \quad \text{or} \quad k\sin\alpha = cte$$

Snell's Law (or Snell-Descartes formula)

As the water depth h decreases,

- C, sin α, and α decrease, and the wave orthogonals become ⊥to the topography (isolines)
- cos α increases and the wave orthogonals move apart

Geometrical refraction

3. Energy relationship: conservation of energy flux between wave orthogonals (extension of shoaling analysis)

$$F_E = EC_g b = c \longrightarrow C_g H^2 b = c$$

such that now:

$$\frac{H}{H_0} = \sqrt{\frac{C_{g0}}{C_g}}\sqrt{\frac{b_0}{b}} = \sqrt{\frac{C_{g0}}{C_g}}\sqrt{\frac{\cos\alpha_0}{\cos\alpha}} = K_s K_r$$

where:

$$K_s = \sqrt{\frac{C_{g0}}{C_g}} \longleftarrow \textbf{Shoaling coefficient}$$

$$K_r = \sqrt{\frac{b_0}{b}} = \sqrt{\frac{\cos\alpha_0}{\cos\alpha}} \longleftarrow \textbf{Refraction coefficient}$$

Geometrical refraction

Rewrite refraction coefficient as:

$$K_r = \left(\frac{1 - \tanh^2(kh)}{\cos^2 \alpha_0} + \tanh^2(kh) \right)^{-1/4} = \left(\frac{1}{\cos^2 \alpha_0} - \tanh^2(kh) \tan^2 \alpha_0 \right)^{-1/4}$$

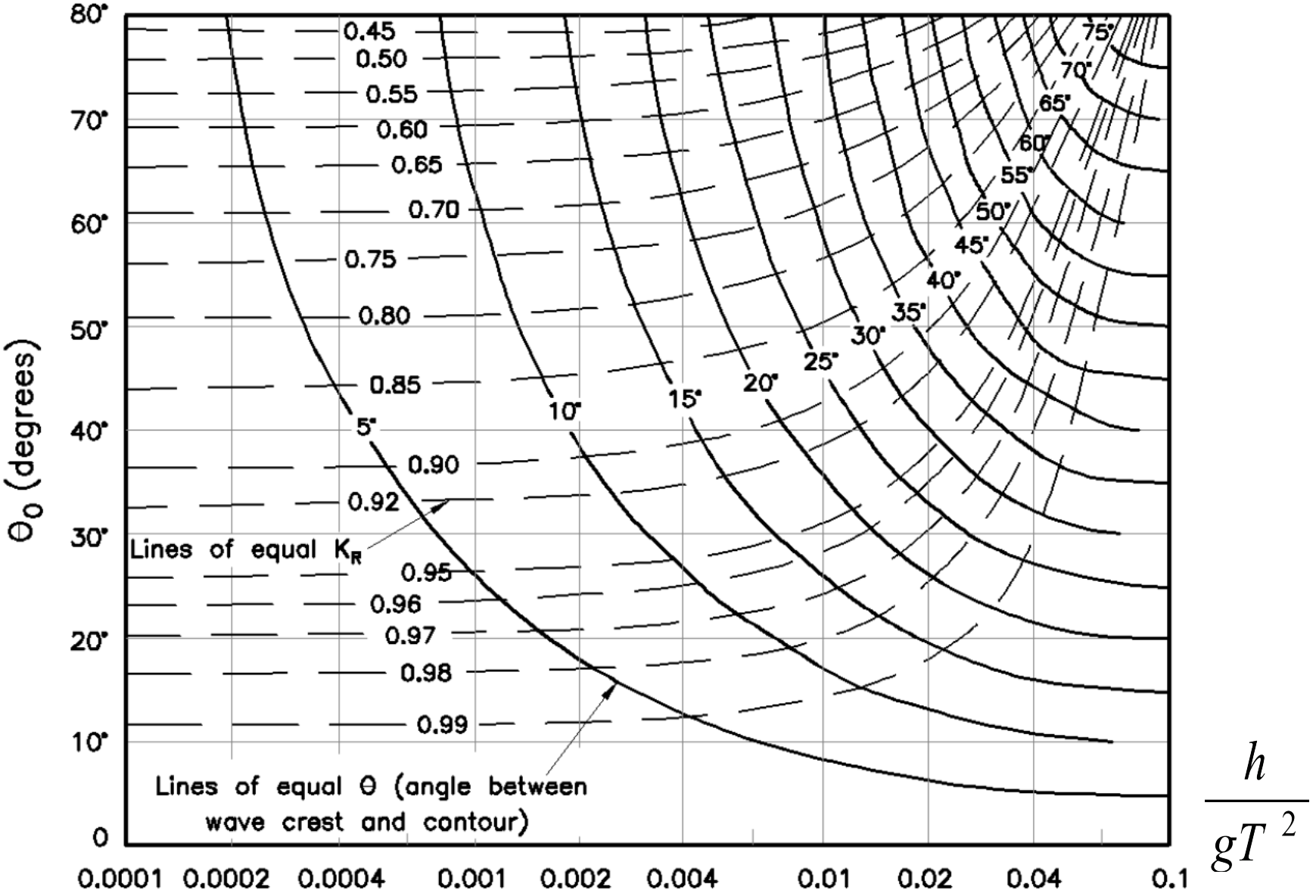
depends on: - relative water depth kh
- offshore wave direction α_0

As the water depth h decreases

- kh decreases, and Kr also decreases (in shallower water, H decreases faster)
- Kr does not tend to 0, but reaches a limit at $\sqrt{\cos \alpha_0}$
- the larger the value of α_0, the lower Kr

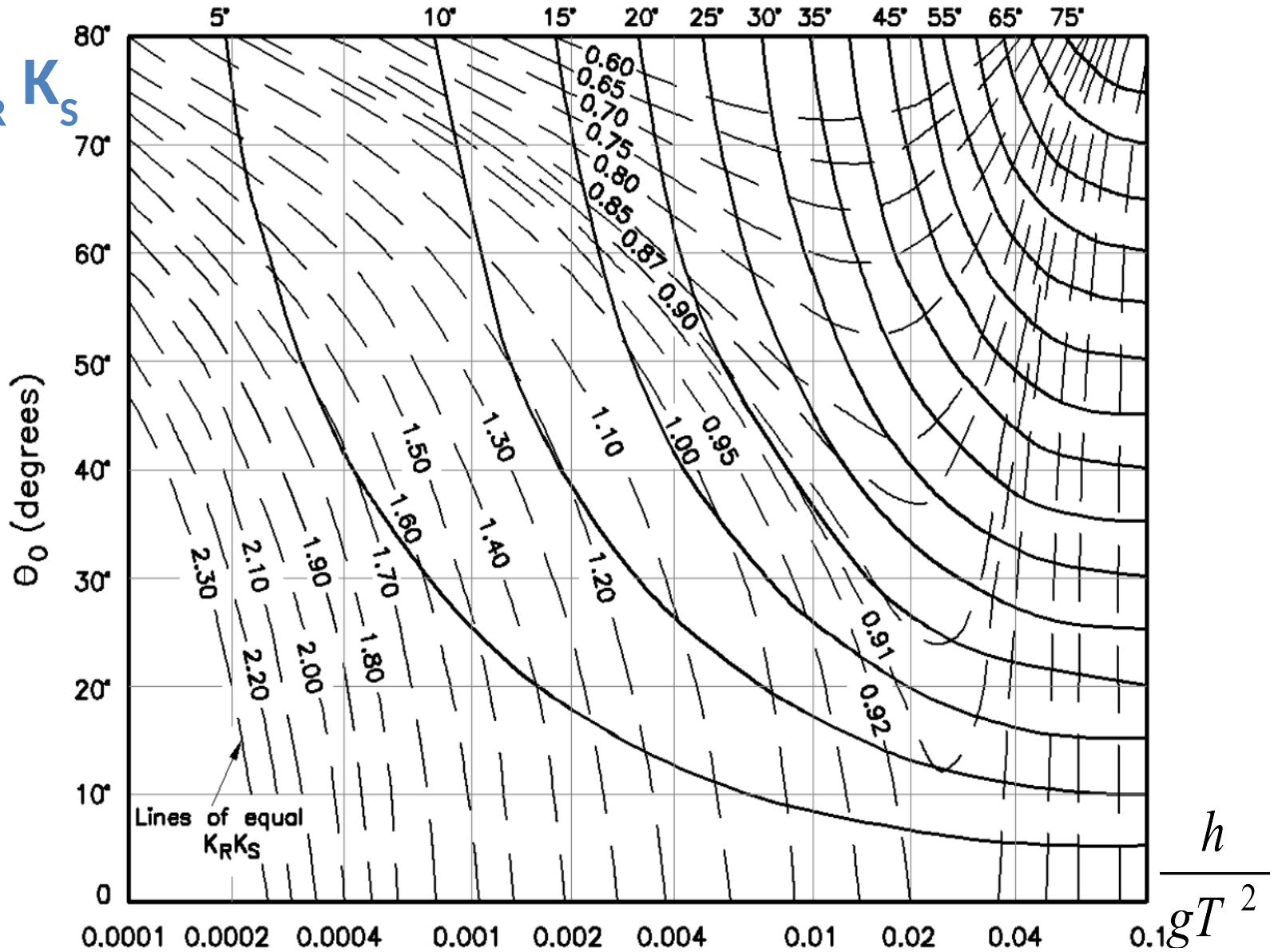
Refraction has a more important role for larger wave incidence angles

Geometrical refraction diagram

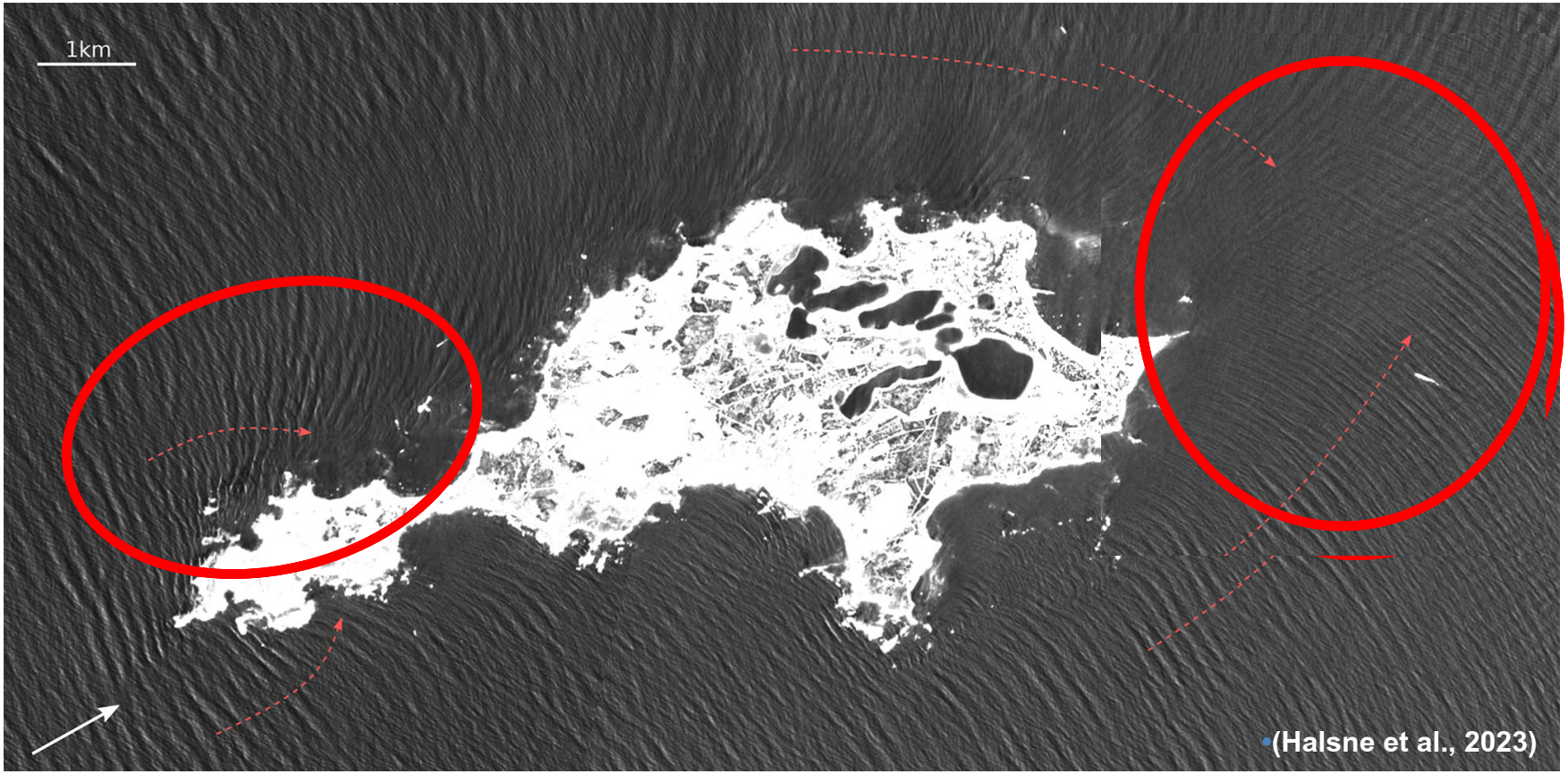


Combined refraction + shoaling diagram

$K_{RS}=K_R K_S$

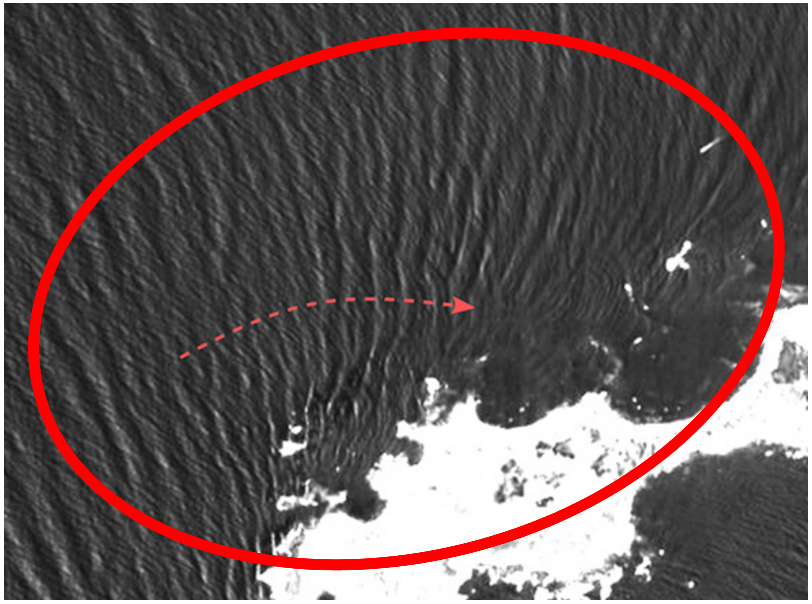


Realistic bathymetry



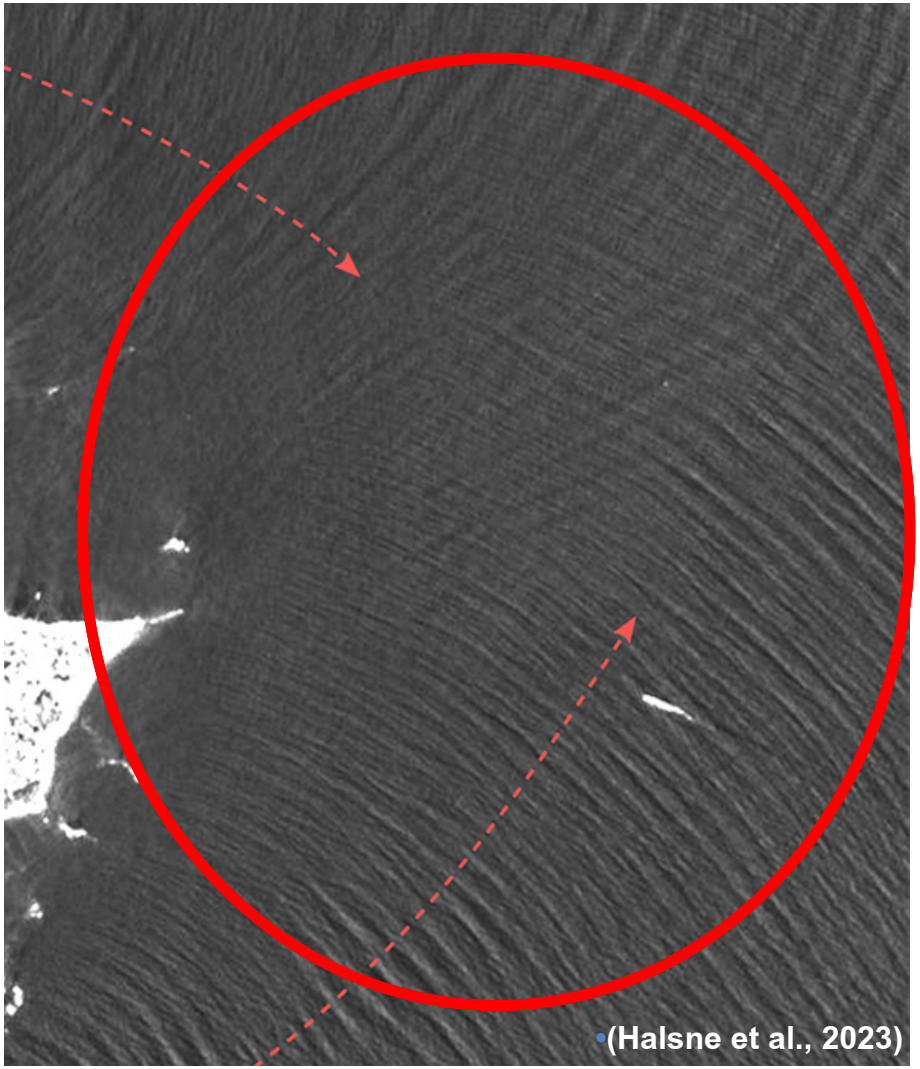
« Depth refraction of swell against Rottnest Island off the coast of Western Australia depicted by the Copernicus Sentinel-2 mission, processed by ESA, in December 2021 »

Realistic bathymetry



How to represent wave refraction over a complex (real) bathymetry ?

Model wave orthogonals



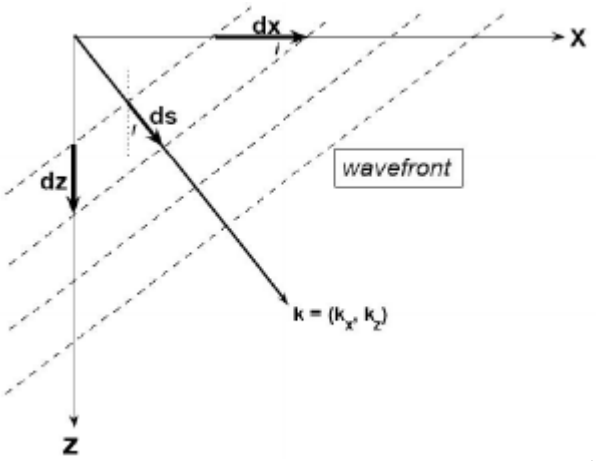
Realistic bathymetry

- Previously linear monotonic decrease in water depth
- Over complex bathymetry: compute wave rays or orthogonals

$$\frac{dx}{ds} = \cos\alpha$$

$$\frac{dy}{ds} = \sin\alpha$$

$$\frac{d\alpha}{ds} = \frac{1}{C}\frac{\partial C}{\partial h}\left[\frac{\partial h}{\partial x}\sin\alpha - \frac{\partial h}{\partial y}\cos\alpha\right]$$

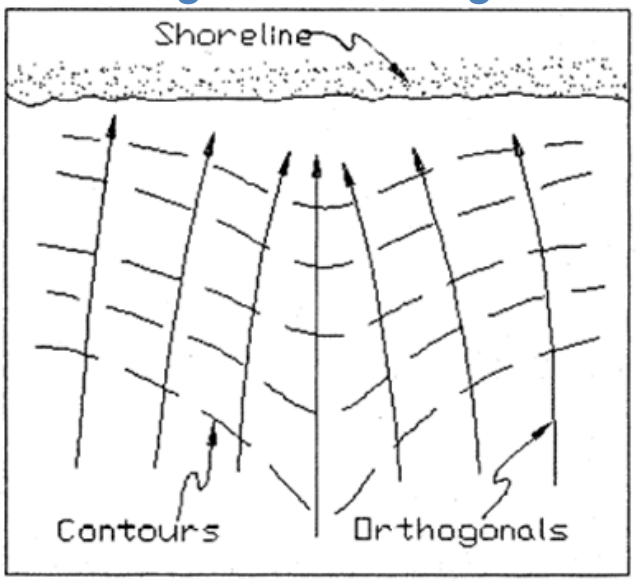


(MIT OpenCoursew are 12.510)

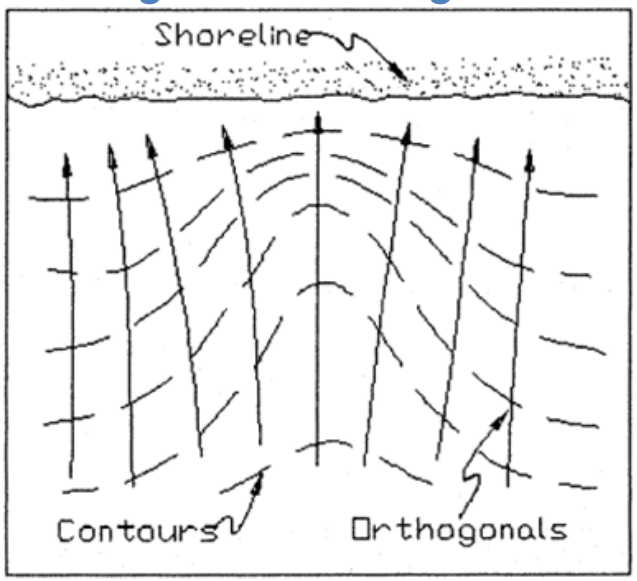
- Speed of a wave packet: $\frac{ds}{dt} = C_g$

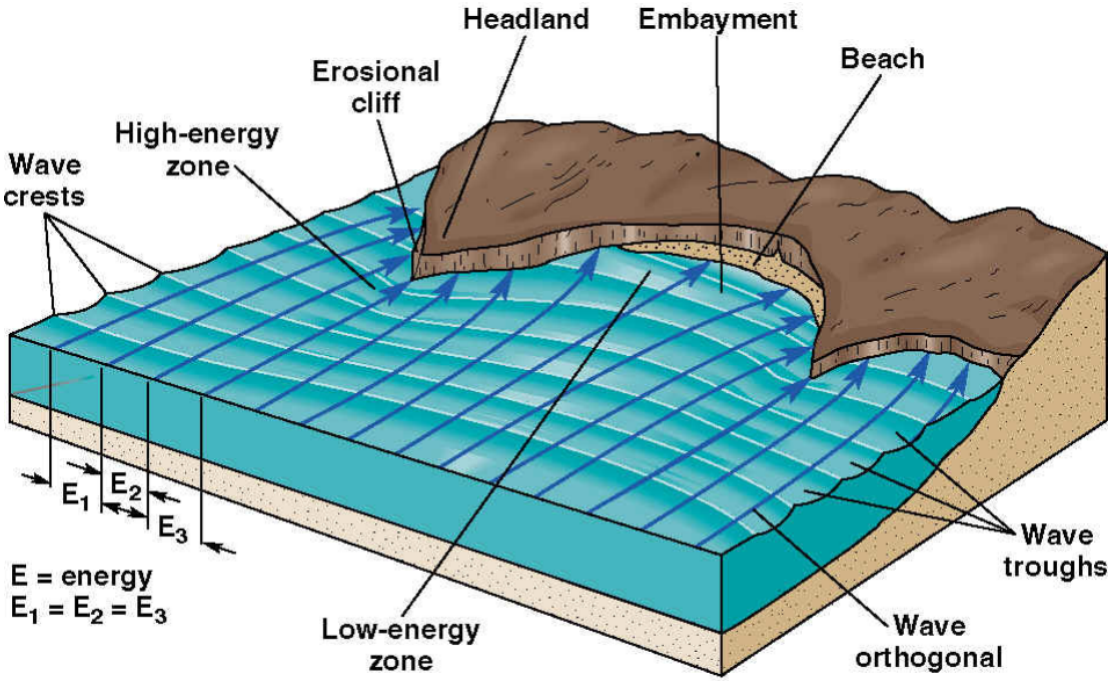
Build wave diagrams or develop a model

Cape, headland
convergence of orthogonals



Bay, canyon
divergence of orthogonals





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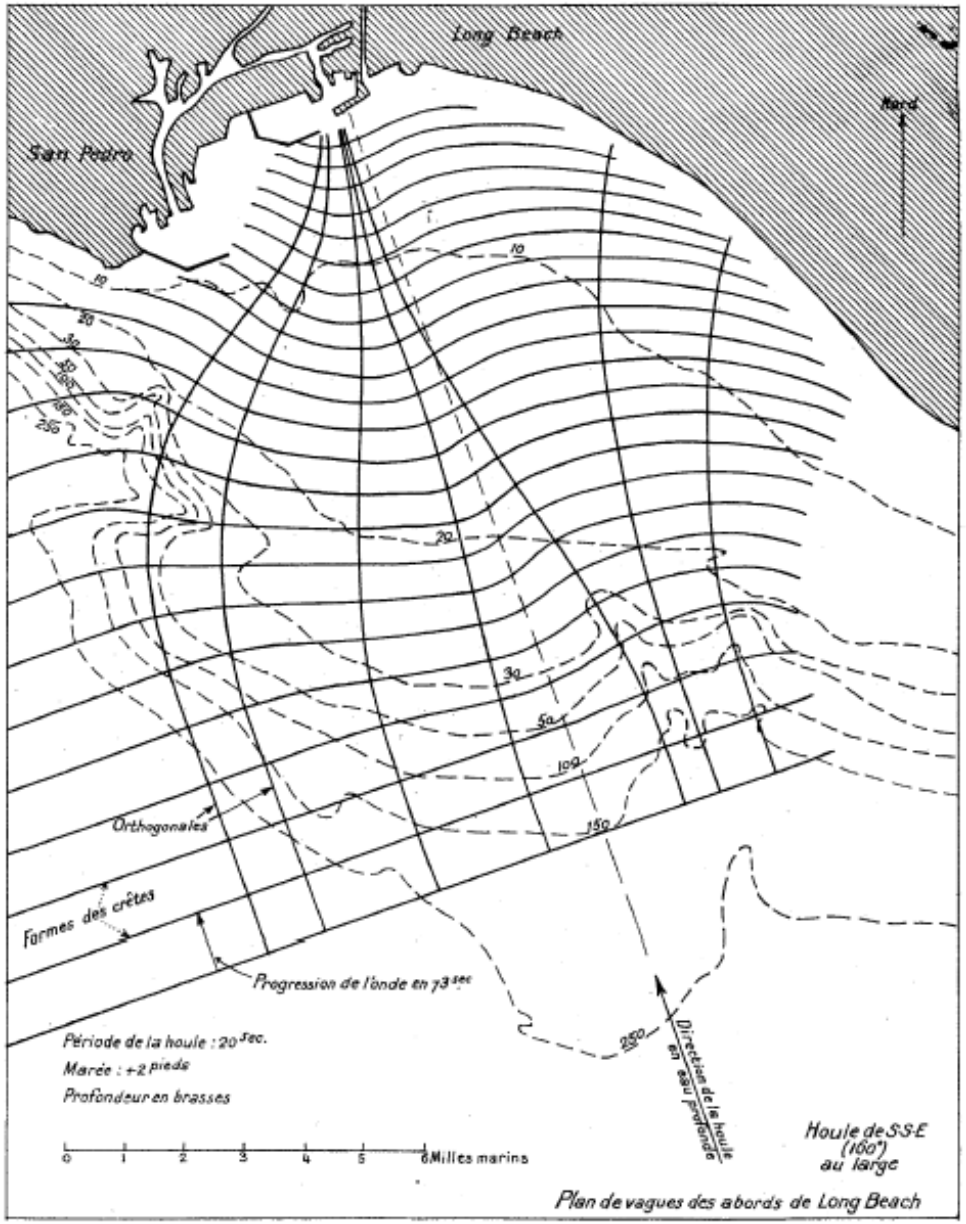
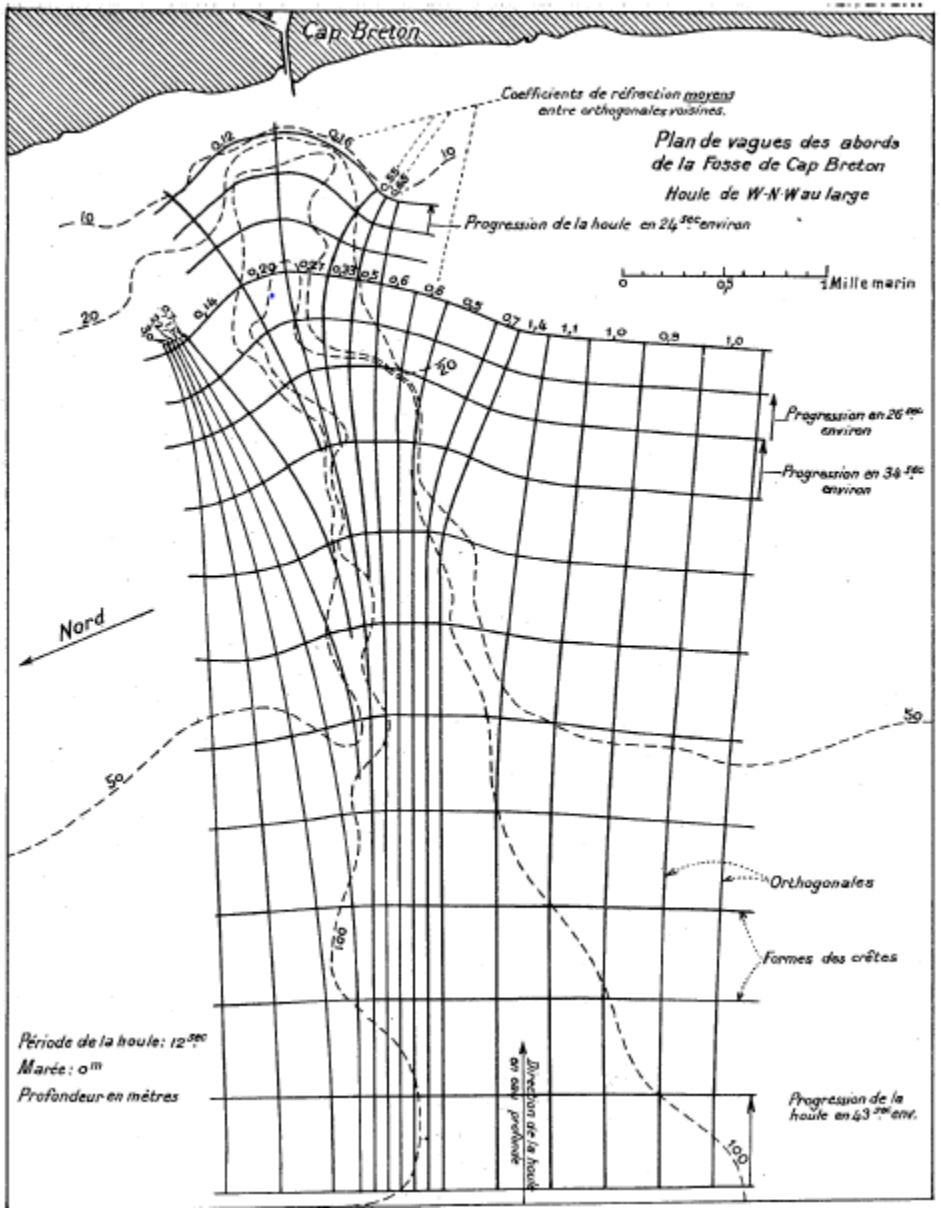
Examples of wave refraction diagrams

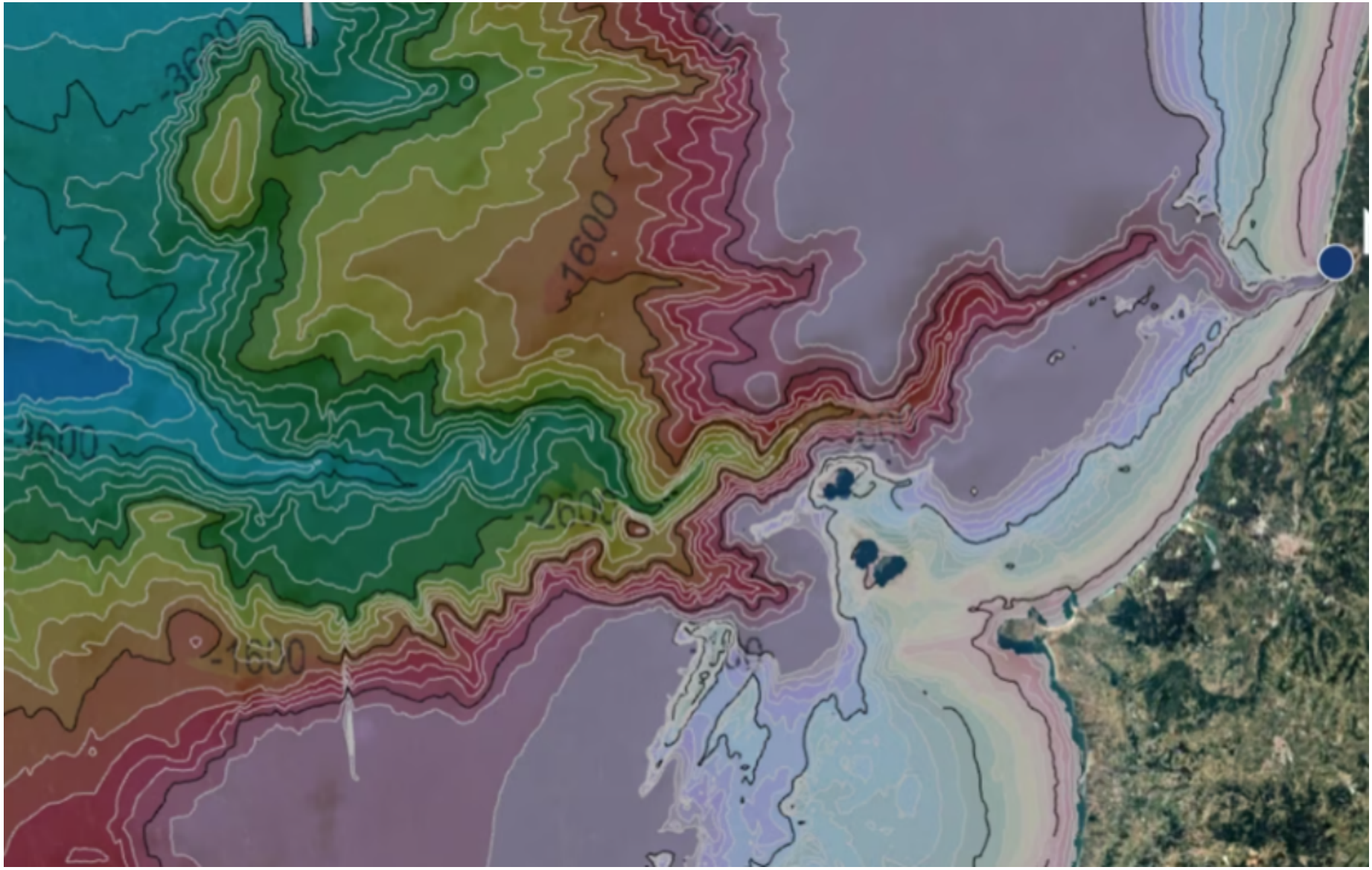


Long Beach (CA, USA)



Cap Breton Gulf (Lacombe, 1950)

Wave refraction diagram



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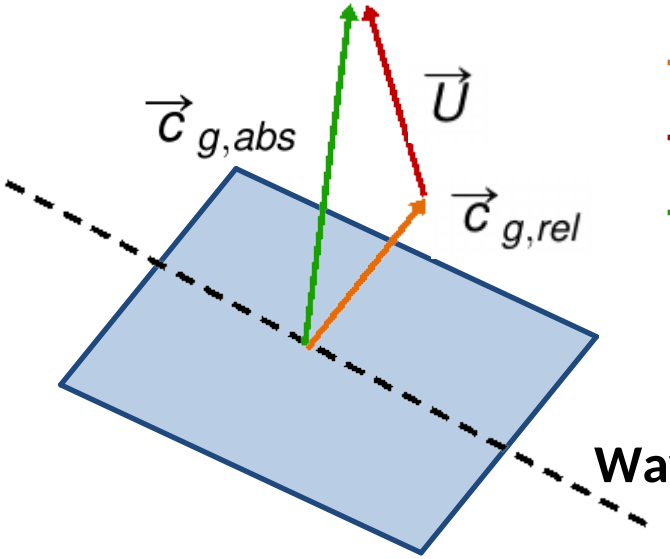
(adapted from Battjes, 1994)

Effects of currents on waves

- Currents (tidal, storm surge, river fluxes, etc.) impact:
  - Wave height (*shoaling*)
  - Wave period/frequency
  - Wave direction (*refraction*)

Effects of currents on waves

- Constant depth, uniform current, linear theory is valid



→ Wave (relative group velocity)

→ Current

→ Energy (absolute group velocity)

(based on Holthuijsen, 2007)

σ	relative frequency
ω	absolute frequency
U_n	horizontal velocity perpendicular to wave crests

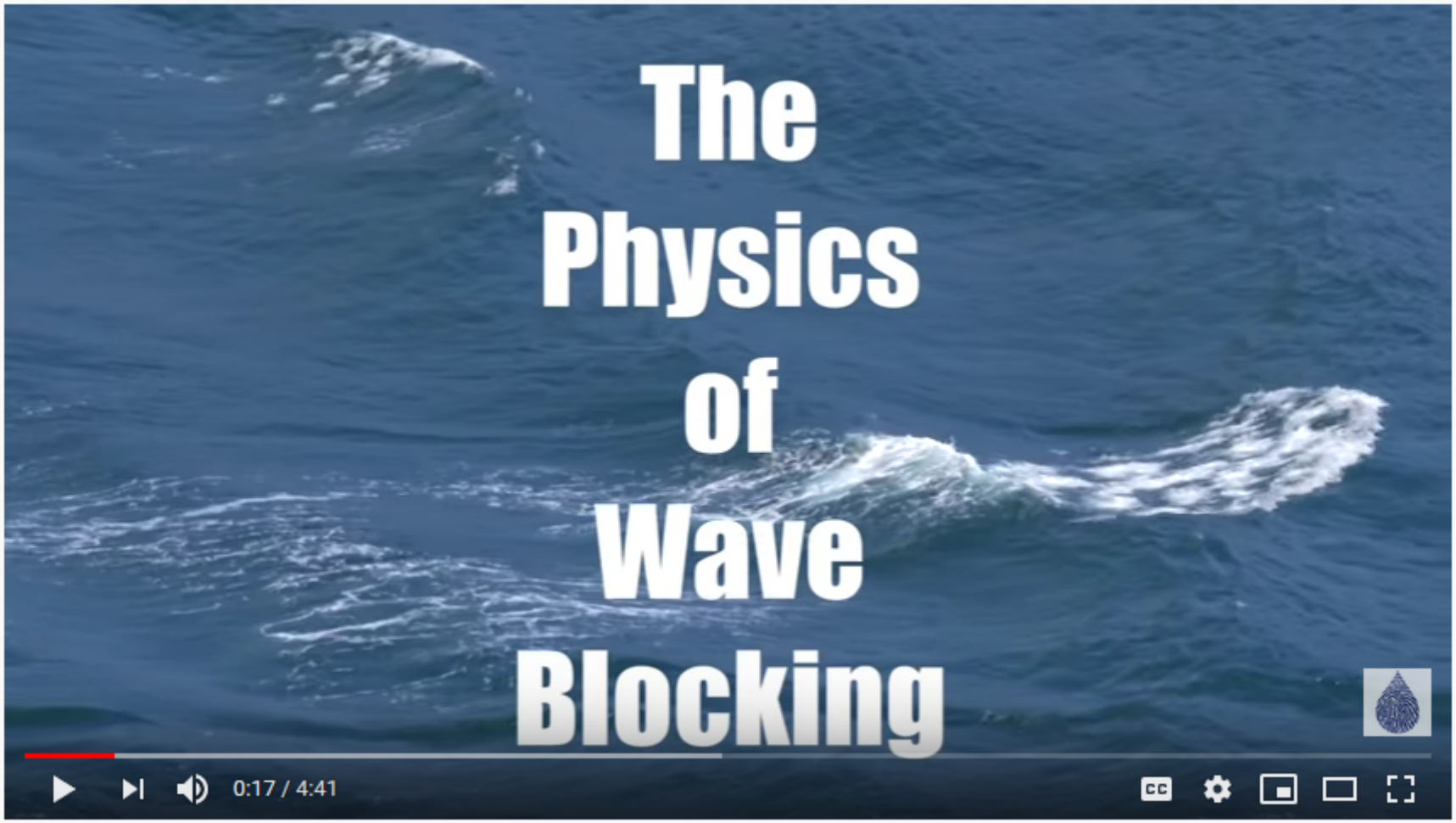
$$\omega^2 = gk\tanh(kh)$$

$$\omega = \sigma + \vec{k}\cdot\vec{U}$$

$$\vec{c}_{g,abs} = \vec{c}_{g,rel} + \vec{U}$$

$$\vec{c}_{abs} = \vec{c}_{rel} + \frac{\vec{k}\cdot\vec{U}}{|k|}$$

Effects of currents on waves



The Physics of Wave Blocking

https://www.youtube.com/watch?v=yVW6XclyBak

Propagation and interactions

Physical processes	Ocean	Continental seas	Coastal zone	Ports
Wind generation	+++	+++	+	
Shoaling		++	+++	++
Bathymetric refraction		++	+++	++
Refraction by currents		+	++	
Diffraction			+	+++
Reflection			+	+++
Triplet interactions			++	+
Quadruplet interactions	+++	+++	+	
Bottom friction		+++	++	
White capping	+++	+++	+	
Bathymetric breaking		+	+++	

(adapted from Battjes, 1994)

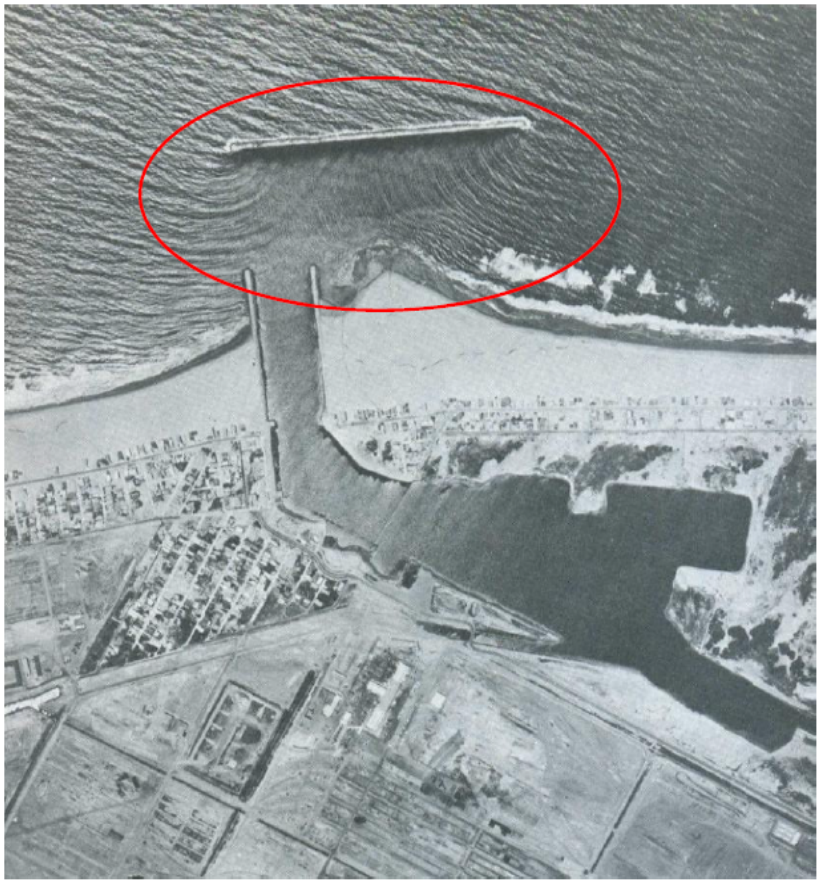
Wave reflection and diffraction

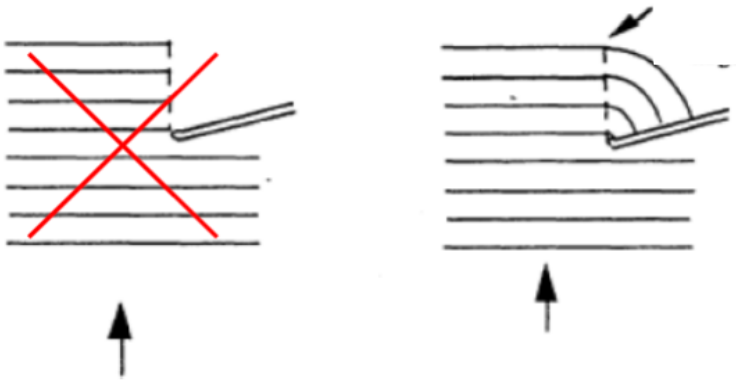
https://www.youtube.com/watch?v=7UhpWR0_rrE

Diffraction around coastal structures

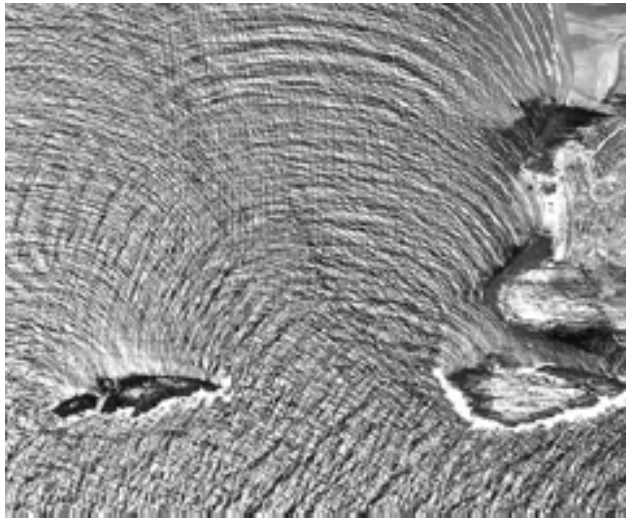
Physical mechanism:

- Transfer of energy to « shadow » zones via wave orthogonals
- Important near obstacles (submerged and emerged) such as breakwaters





Diffraction: linear theory (flat bottom)



Assumptions:

- Inviscid fluid, irrotationnal flow
- Linear waves (H/L<<1, H/h<<1)
- Flat bottom

Equations :

- Velocity potenial from Airy wave theory:

$$\Phi(x, y, z, t) = -i\frac{gH}{2\omega}\frac{\cosh\left(k(z+h)\right)}{\cosh\left(kh\right)}e^{-i\omega t}\,\phi(x, y)$$

- Laplace's equation implies (Helmholtz equation):

$$\Delta\phi + k^2\phi = 0$$

- Boundary condition at vertical boundaries: $\dfrac{\partial\phi}{\partial n} = 0$

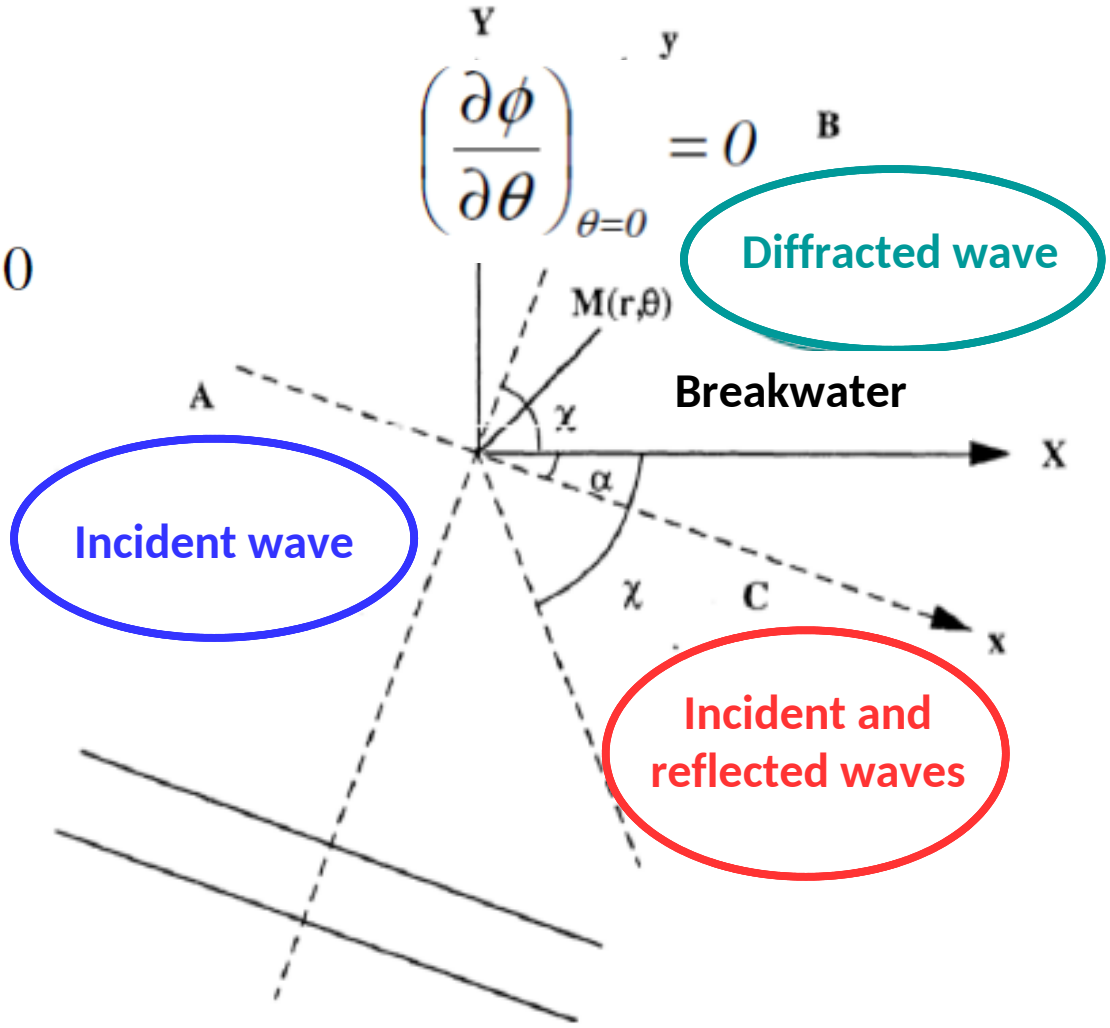
Diffraction : semi-infinite breakwater

Transformation to polar coordinates :

$$\frac{\partial^2\phi}{\partial r^2}+\frac{1}{r}\frac{\partial\phi}{\partial r}+\frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2}+k^2\phi=0$$

Analytical solution

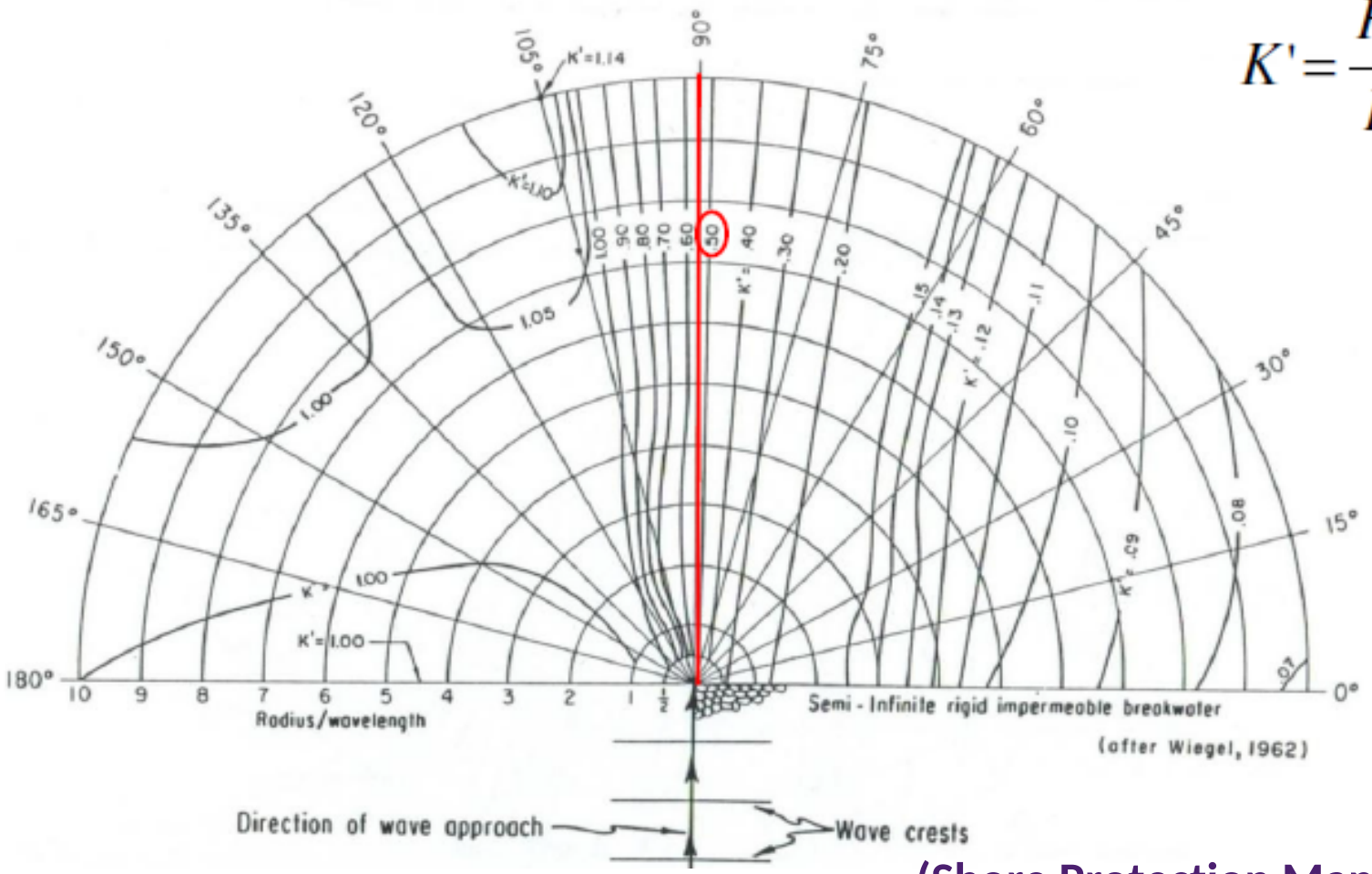
(Putnam & Arthur, 1948)



Diffraction : semi-infinite breakwater

Diffraction diagram for a semi-infinite breakwater with an incident wave angle of 90°

$$K' = \frac{H_{diff}}{H_{inc}}$$

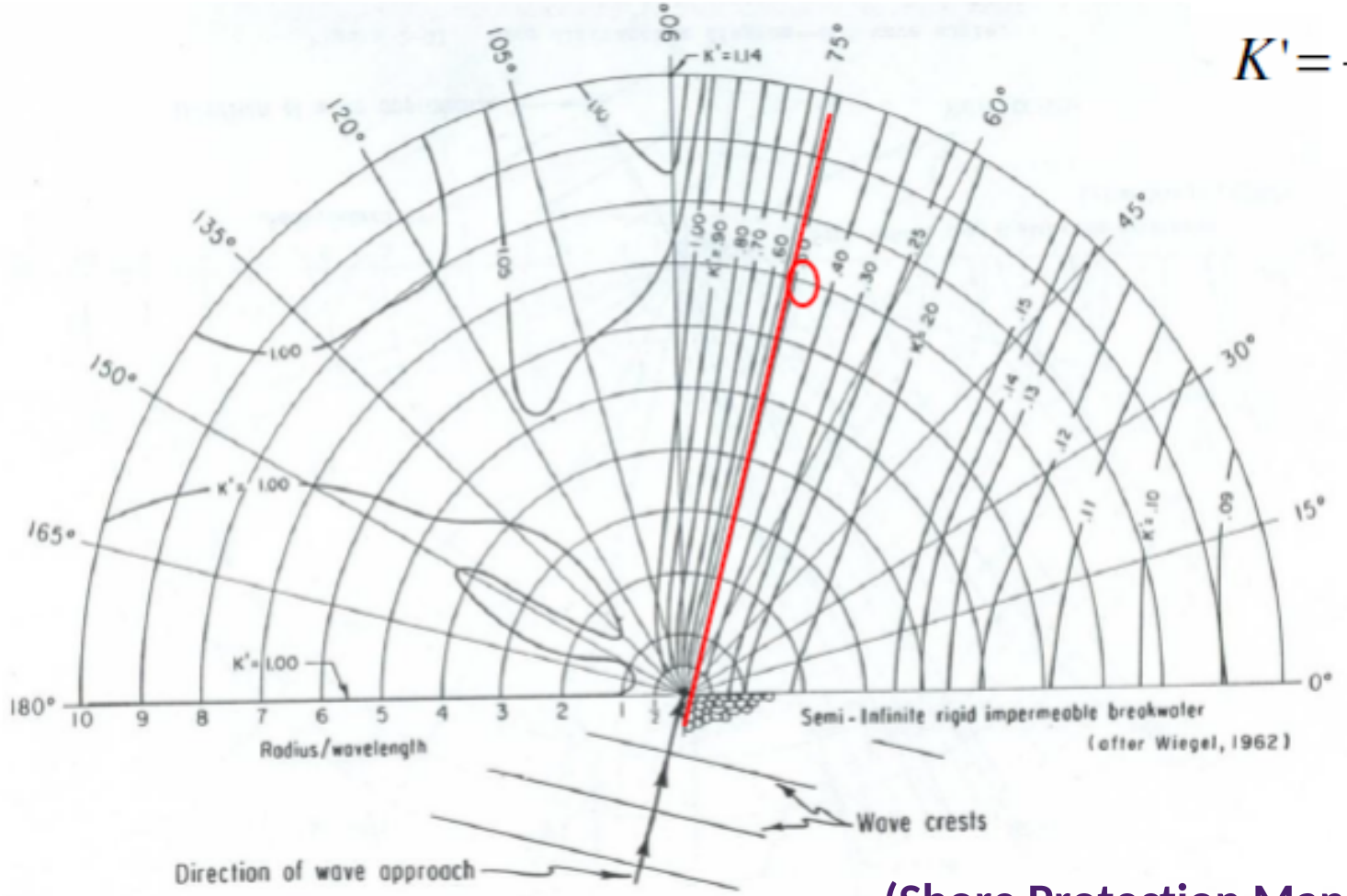


(Shore Protection Manual, CERC)

Diffraction : semi-infinite breakwater

Diffraction diagram for a semi-infinite breakwater with an incident wave angle of 75°

$$K' = \frac{H_{diff}}{H_{inc}}$$



(Shore Protection Manual, CERC)

Propagation and interactions

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Refraction by currents		+	++	
Diffraction			+	+++
Reflection			+	+++
Triplet interactions			++	+
Quadruplet interactions	+++	+++	+	
Bottom friction		+++	++	
White capping	+++	+++	+	
Bathymetric breaking		+	+++	

(adapted from Battjes, 1994)

Reflection from coastal structures

Physical mechanism:

- Wave energy is reflected, dissipated, or transmitted (in proportions that depend on the structure's characteristics)
- Important for wave propagation in ports, navigation, stability of coastal structures, scouring
- Goal is to design coastal structures that maximize dissipation

Reflection from coastal structures

Reflection coefficient:

$$R = \frac{H_{réfl}}{H_{inc}} \qquad 0 \leq R \leq 1$$

Depends on :

- structure's characteristics
(slope, roughness, porosity)

- wave conditions
(steepness, incident angle)

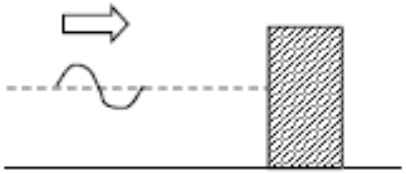
Dissipation

Physical processes	Ocean	Continental seas	Coastal zone	Ports
Wind generation	+++	+++	+	
Shoaling		++	+++	++
Bathymetric refraction		++	+++	++
Refraction by currents		+	++	
Diffraction			+	+++
Reflection			+	+++
Triplet interactions			++	+
Quadruplet interactions	+++	+++	+	
Bottom friction		+++	++	
White capping	+++	+++	+	
Bathymetric breaking		+	+++	

(adapted from Battjes, 1994)

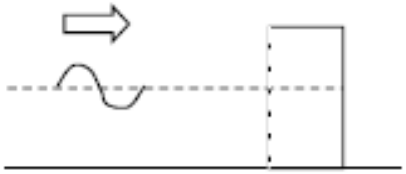
Reflection from coastal structures

- **Solid or porous vertical boundary**



Total reflection:

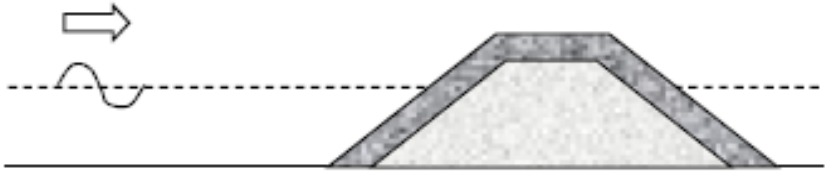
$R \sim 1$



Partial reflection:

$0.4 \leq R \leq 0.8$

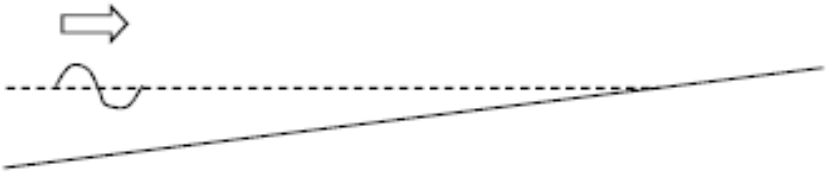
- **Breakwater**



Partial reflection:

$0.2 \leq R \leq 0.5$

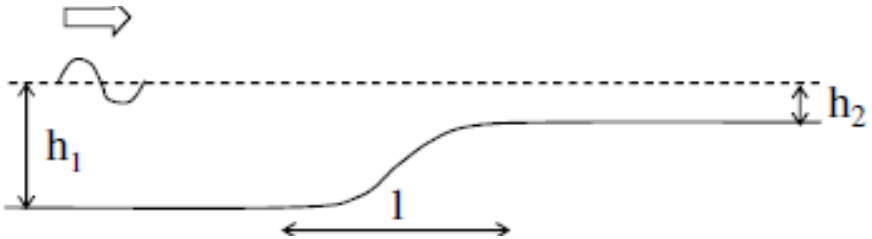
- **Beaches, coastal profiles**



Weak reflection:

$0.05 \leq R \leq 0.2$

- **Bathymetric changes (depth changes, immerged structures)**



Depending on h_1, h_2, and l :

$0.05 \leq R \leq 0.6$

Course schedule

I. Wave dynamics	Project	Lecturer	Date
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	26/09
2. Linear wave theory	Characterizing offshore waves	M. Yates	03/10
3. Nearshore wave propagation	Calculating extreme wave conditions	M. Yates	10/10
4. Coastal hydrodynamics	**Student mini-presentations**	**M. Yates**	**17/10**
II. Numerical wave modeling			
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	24/10
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	07/11
III. Wave-structure interactions			
7. Dynamics of a body in waves	Wave load estimation	J. Harris	14/11
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	28/11
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	05/12
10. Final project presentations 13h30 - 17h15			***19 /12***

Propagation and interactions

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Bottom friction		+++	++	
White capping	+++	+++	+	
Bathymetric breaking		+	+++	

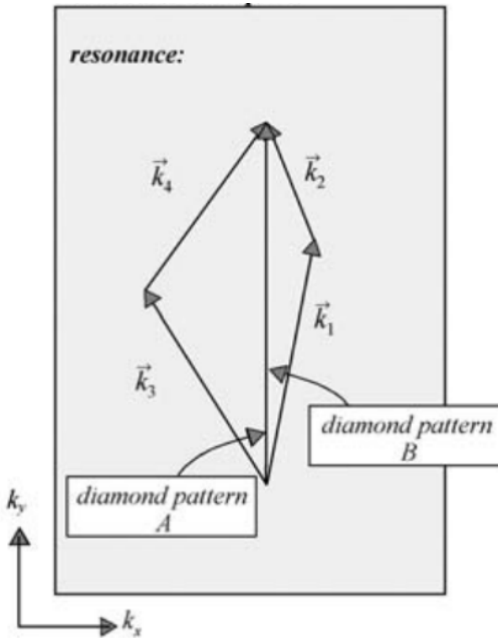
(adapted from Battjes, 1994)

Wave-wave interactions

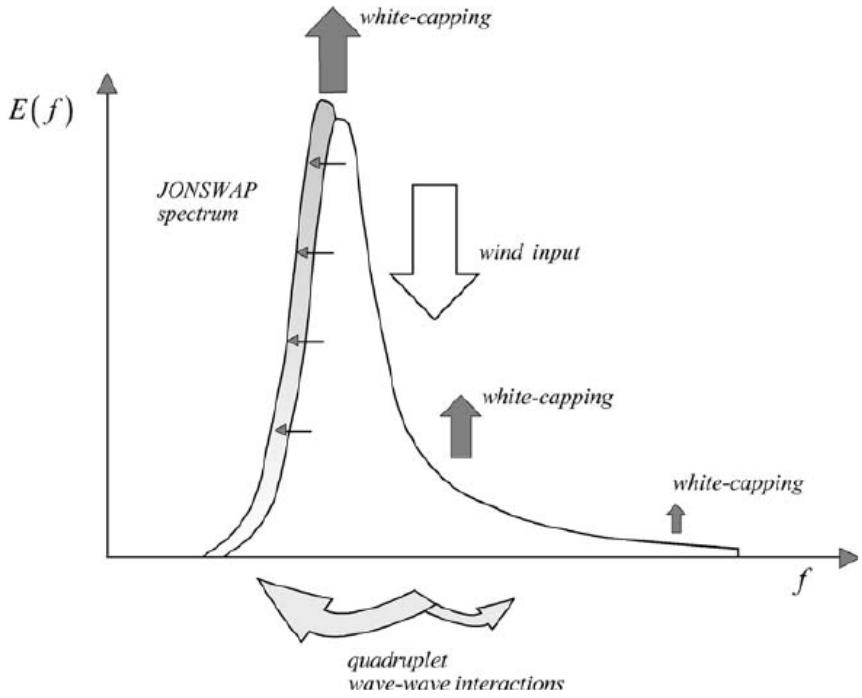
Resonance between 3 or 4 wave frequencies:

- **transfer of energy in wave spectrum between different frequencies**

In deep water, interactions between quadruplets



$$\vec{k_1} + \vec{k_2} = \vec{k_3} + \vec{k_4}$$
$$\omega_1 + \omega_2 = \omega_3 + \omega_4$$



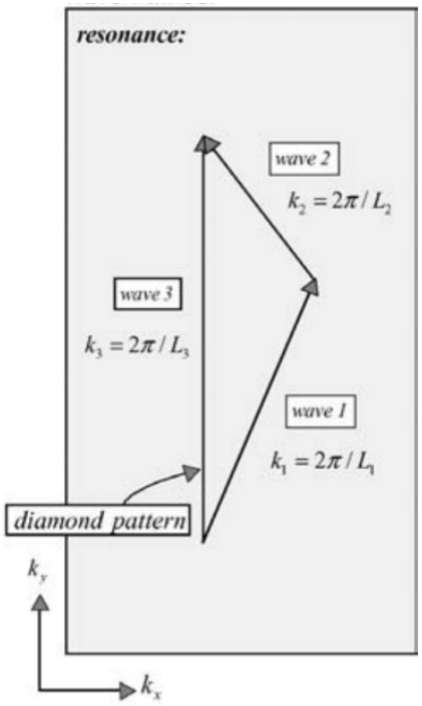
wave energy transfer from the peak to higher and lower frequencies

Wave-wave interactions

Resonance between 3 or 4 wave frequencies:

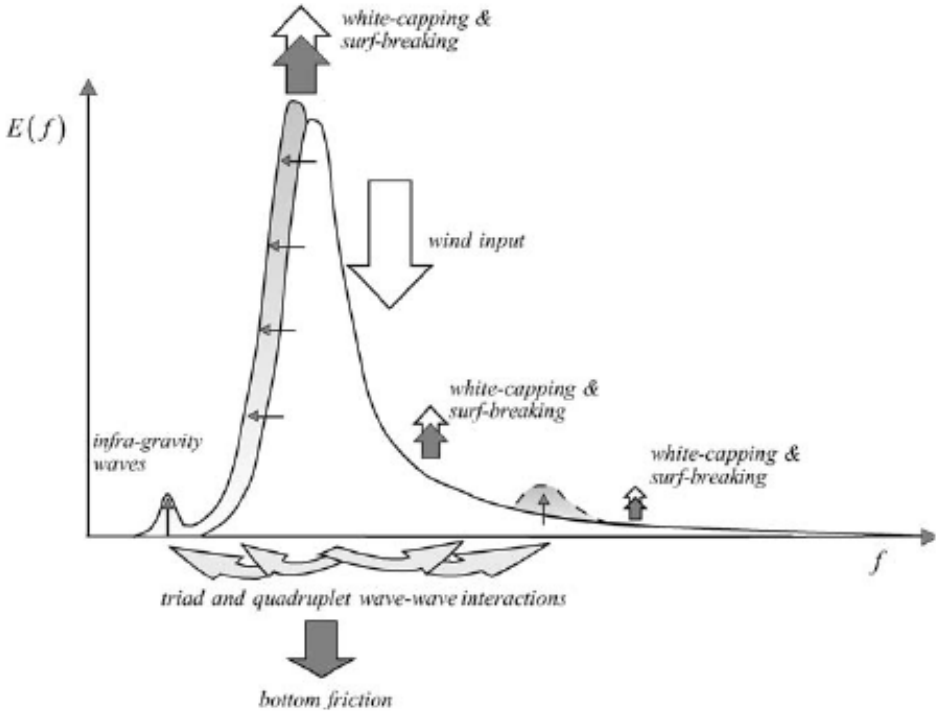
- transfer of energy in wave spectrum between different frequencies

In shallow water, interactions between triplets



$$\vec{k_1} + \vec{k_2} = \vec{k_3}$$

$$\omega_1 + \omega_2 = \omega_3$$



wave energy transfer from the peak to higher and lower frequencies *(in particular between harmonics)*